Exact POMDP Solutions: α-vectors

Recap

Recap

- POMDP

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$

Recap

- POMDP
- Belief Updates

$$(S, A, O, R, T, Z, \gamma)$$

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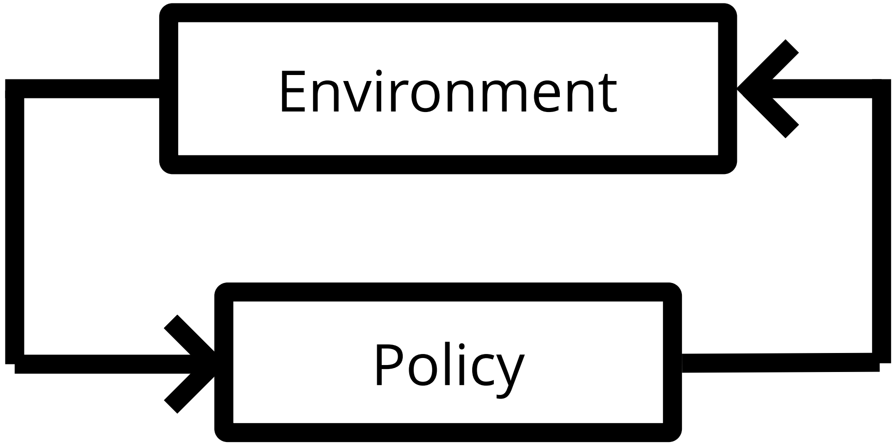
MDP Sense-Plan-Act Loop



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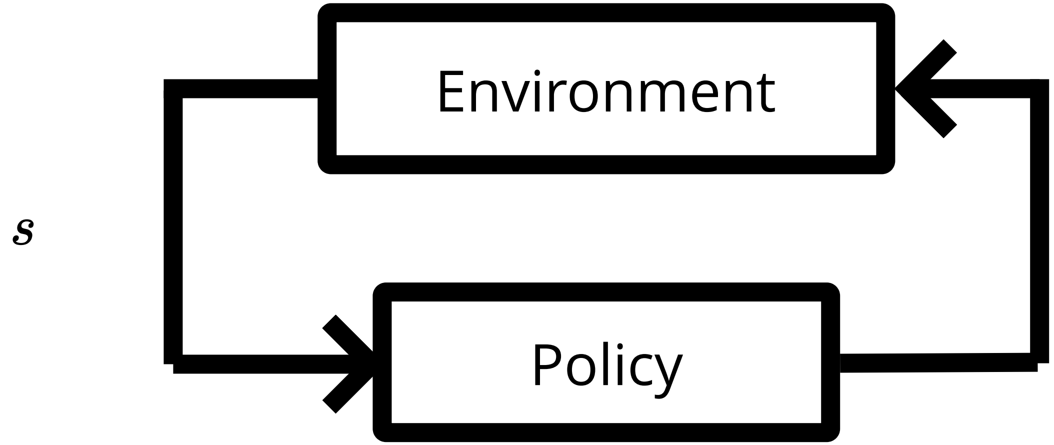

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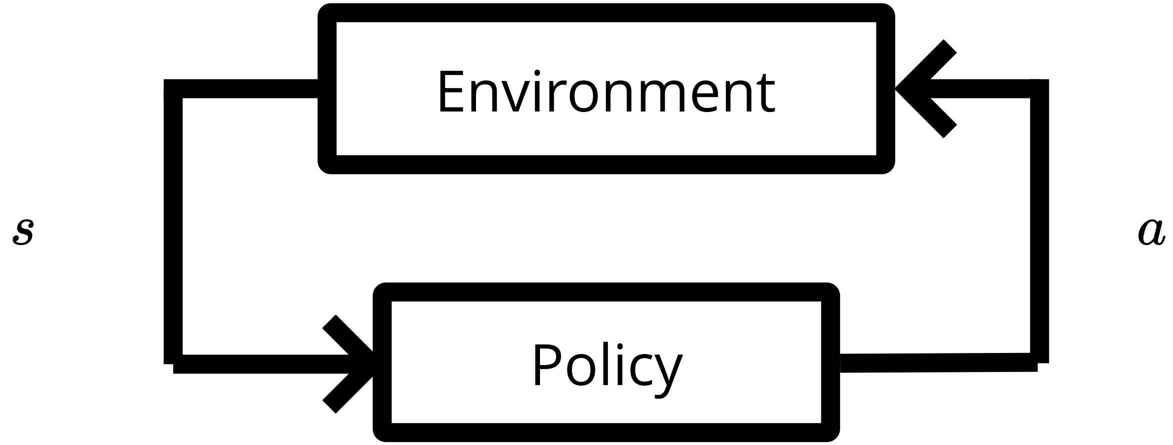
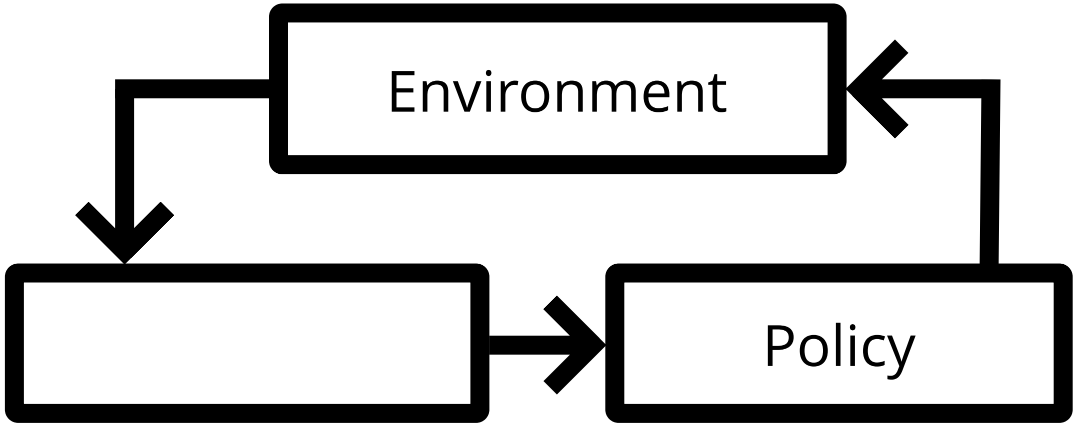


POMDP Sense-Plan-Act Loop



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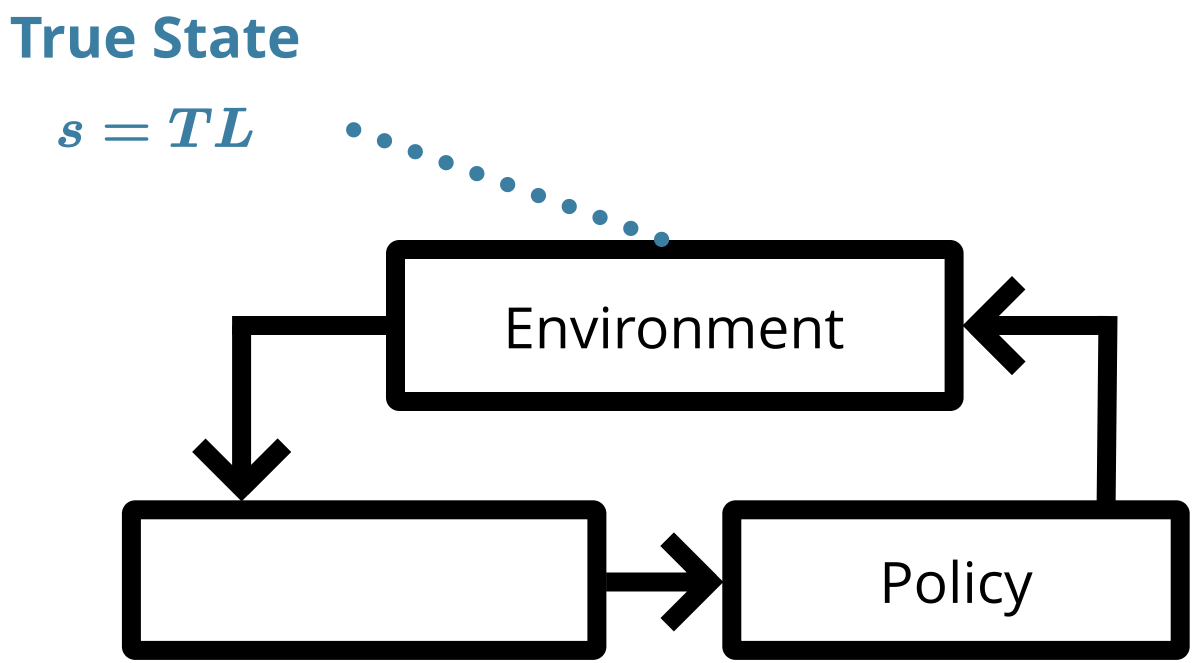

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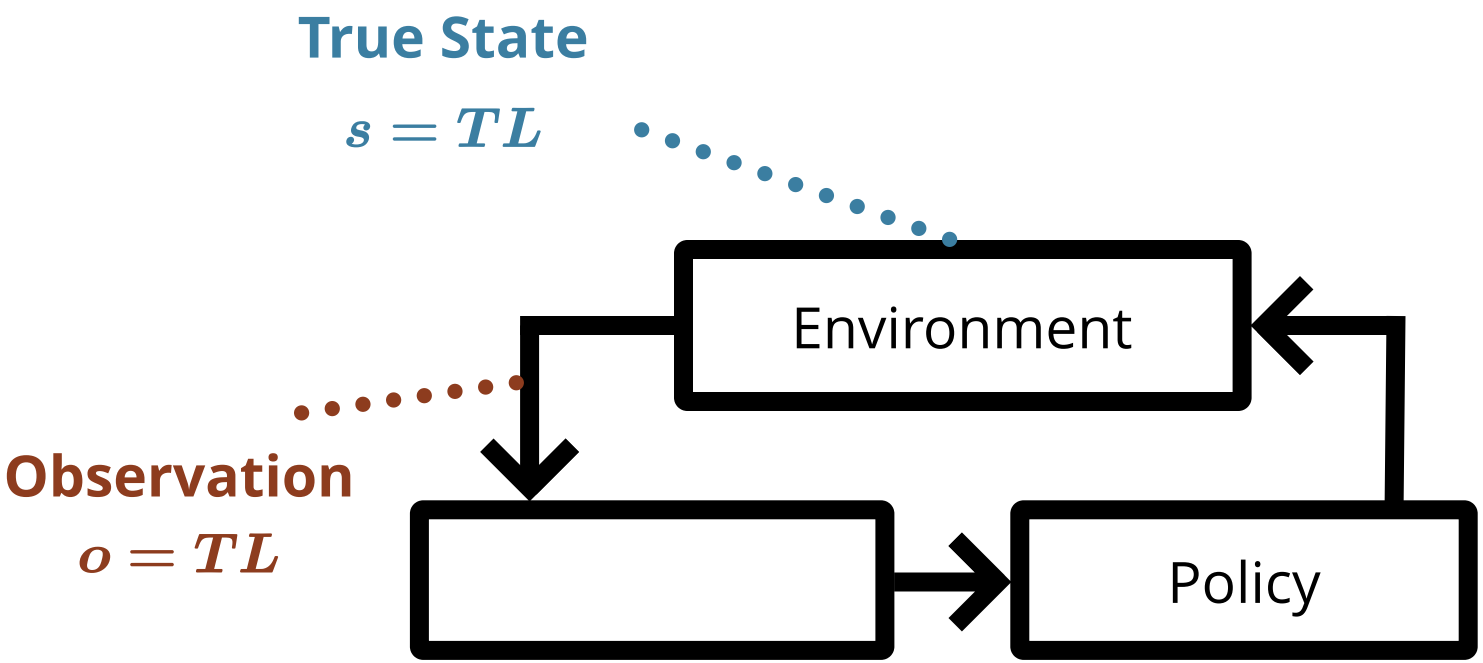

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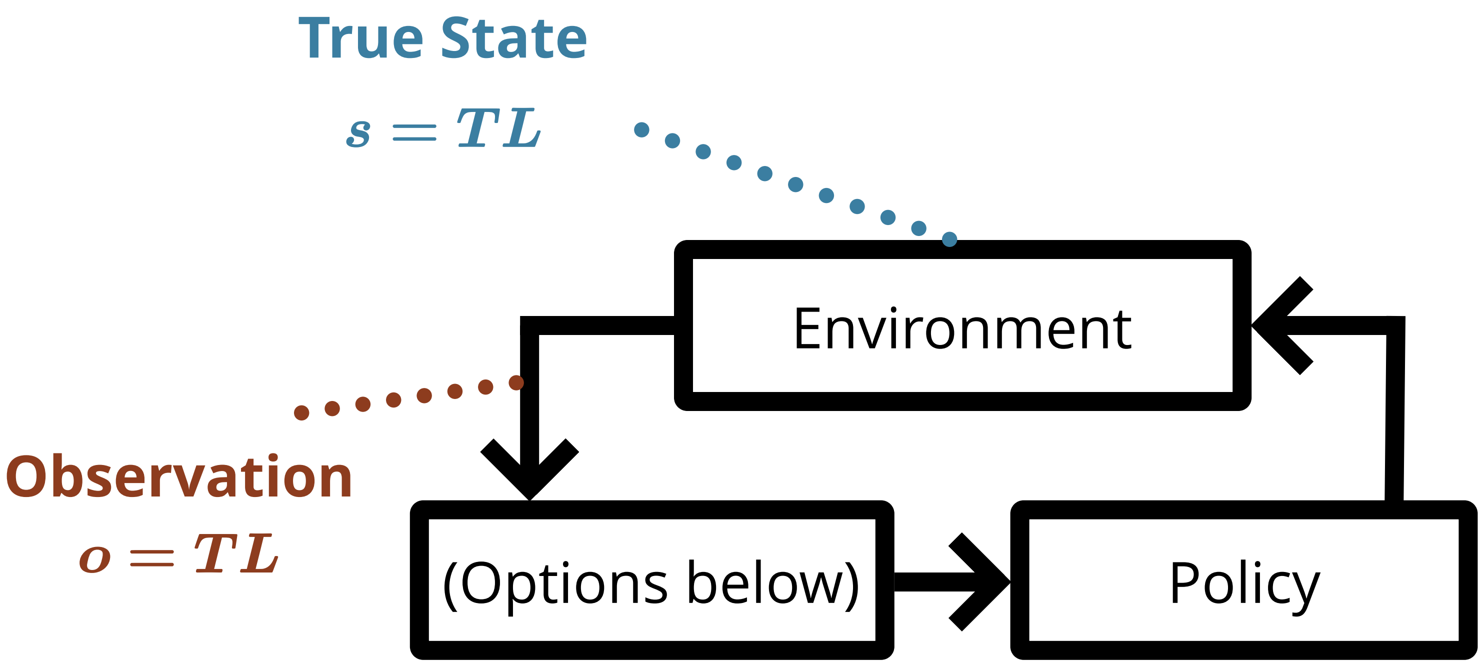

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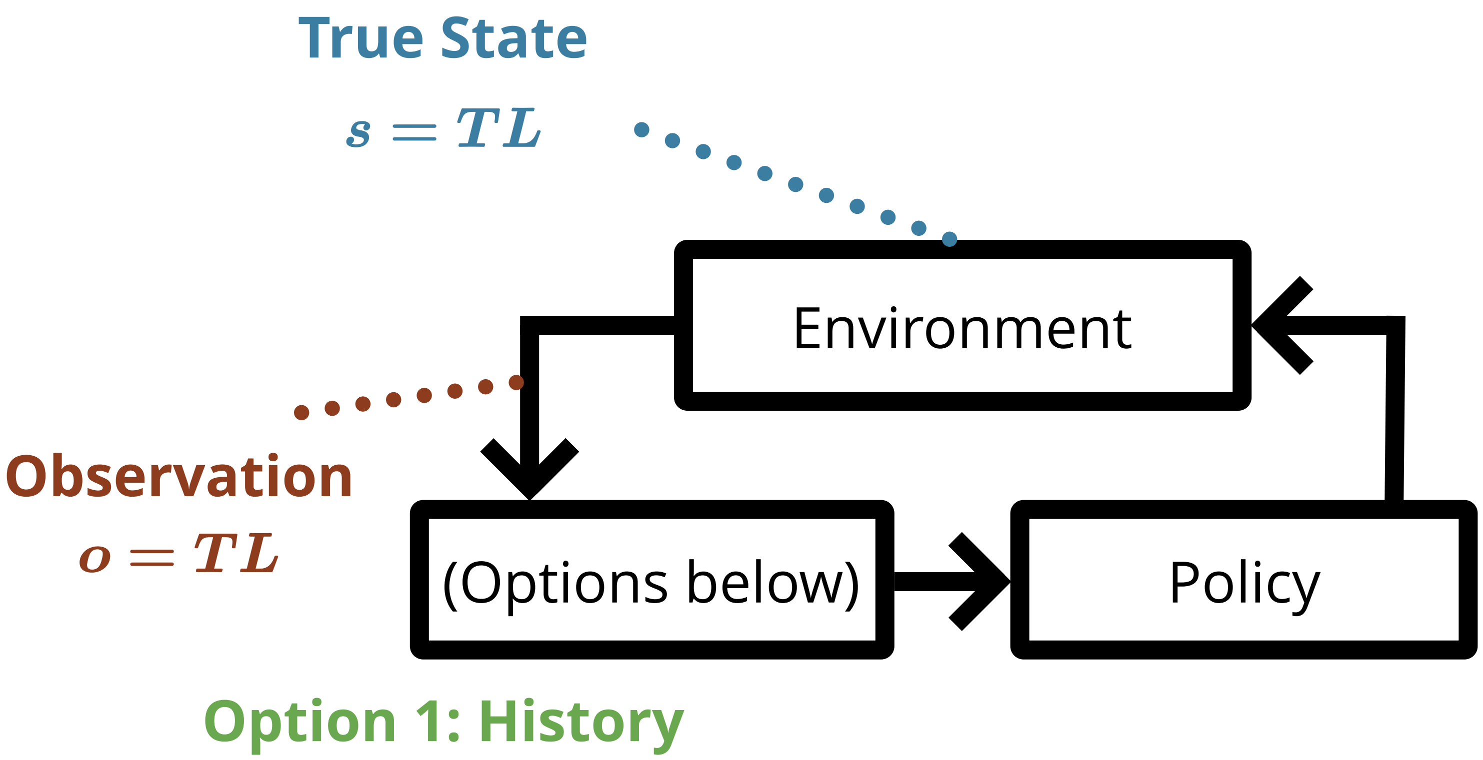

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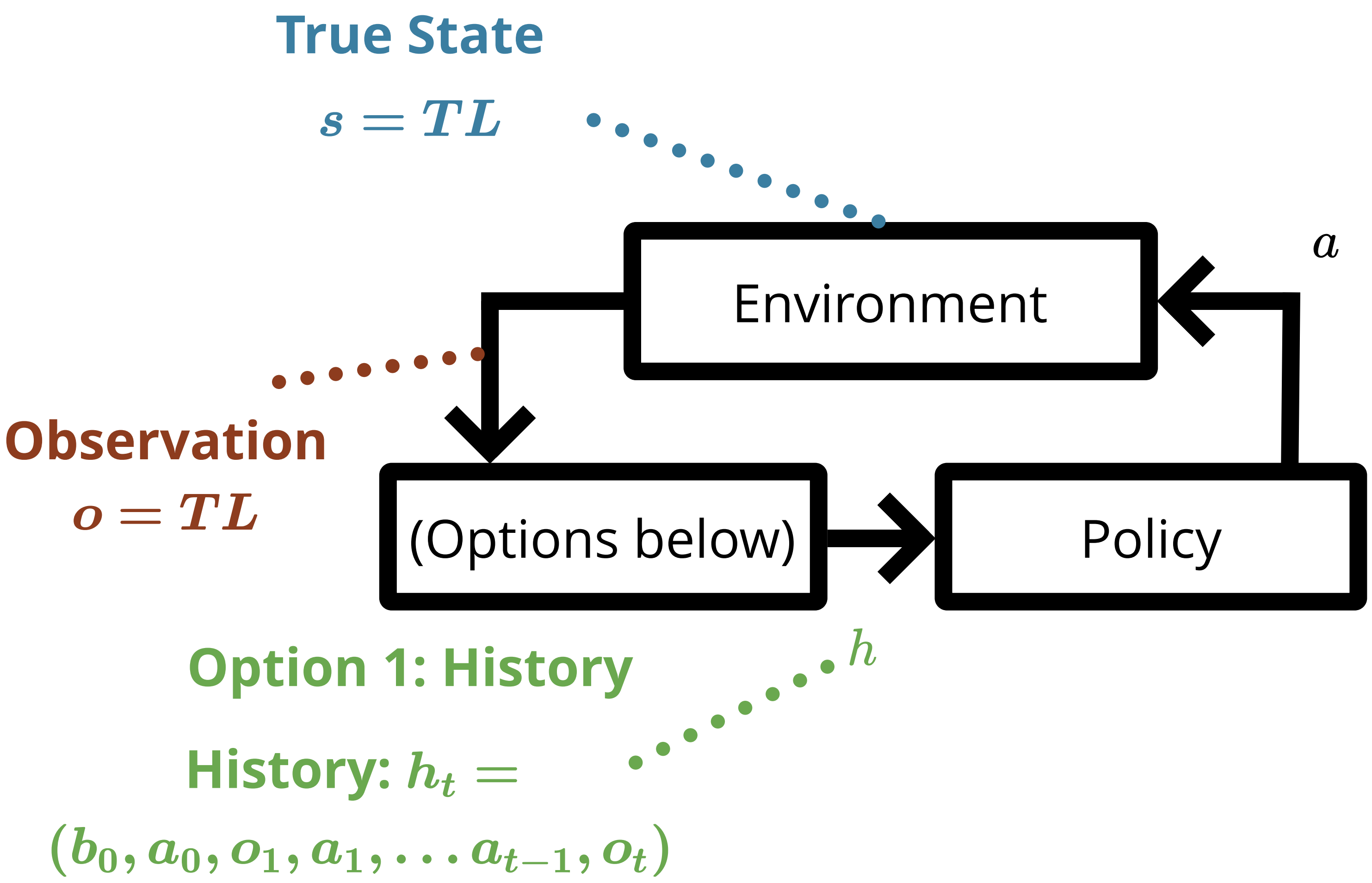

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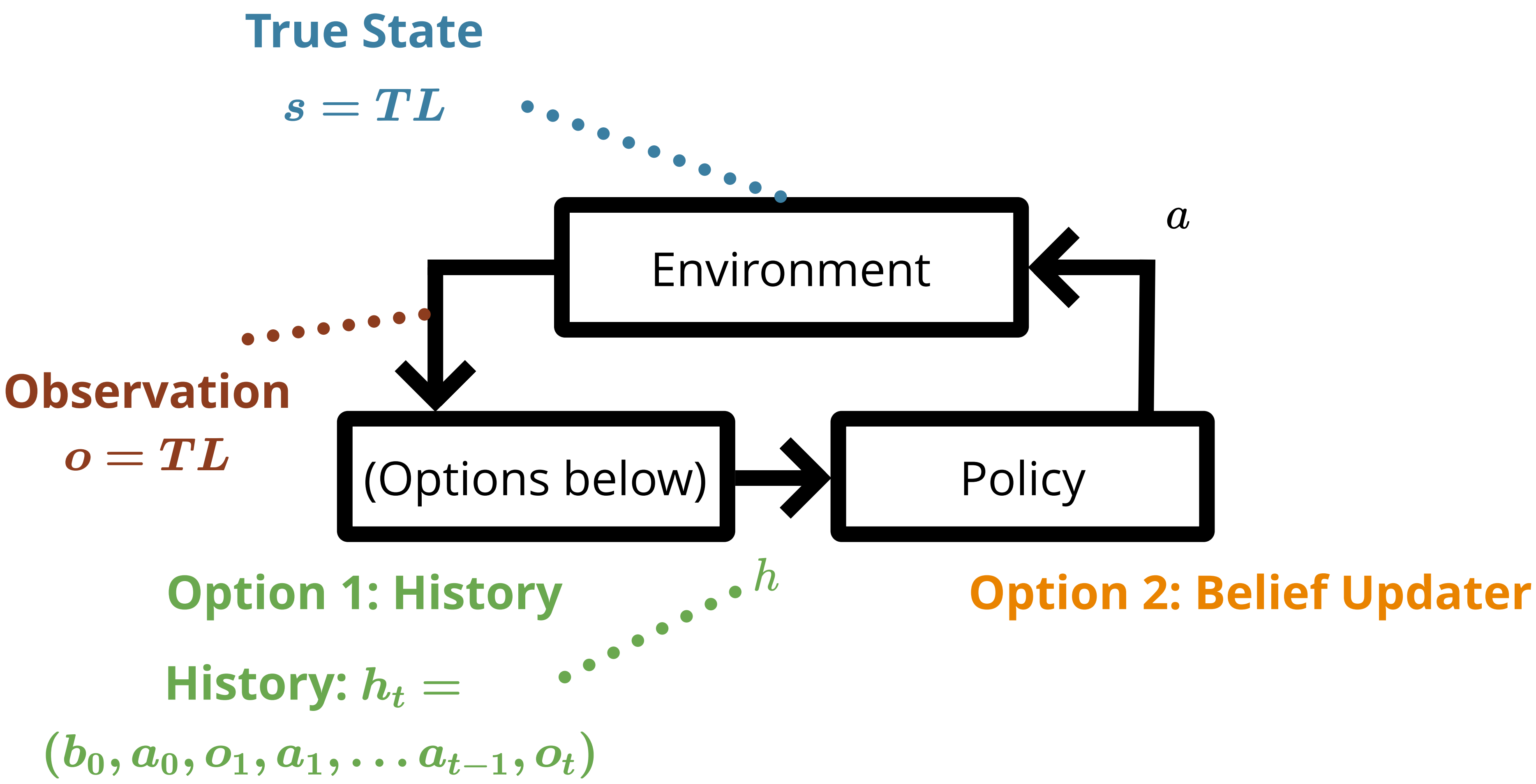

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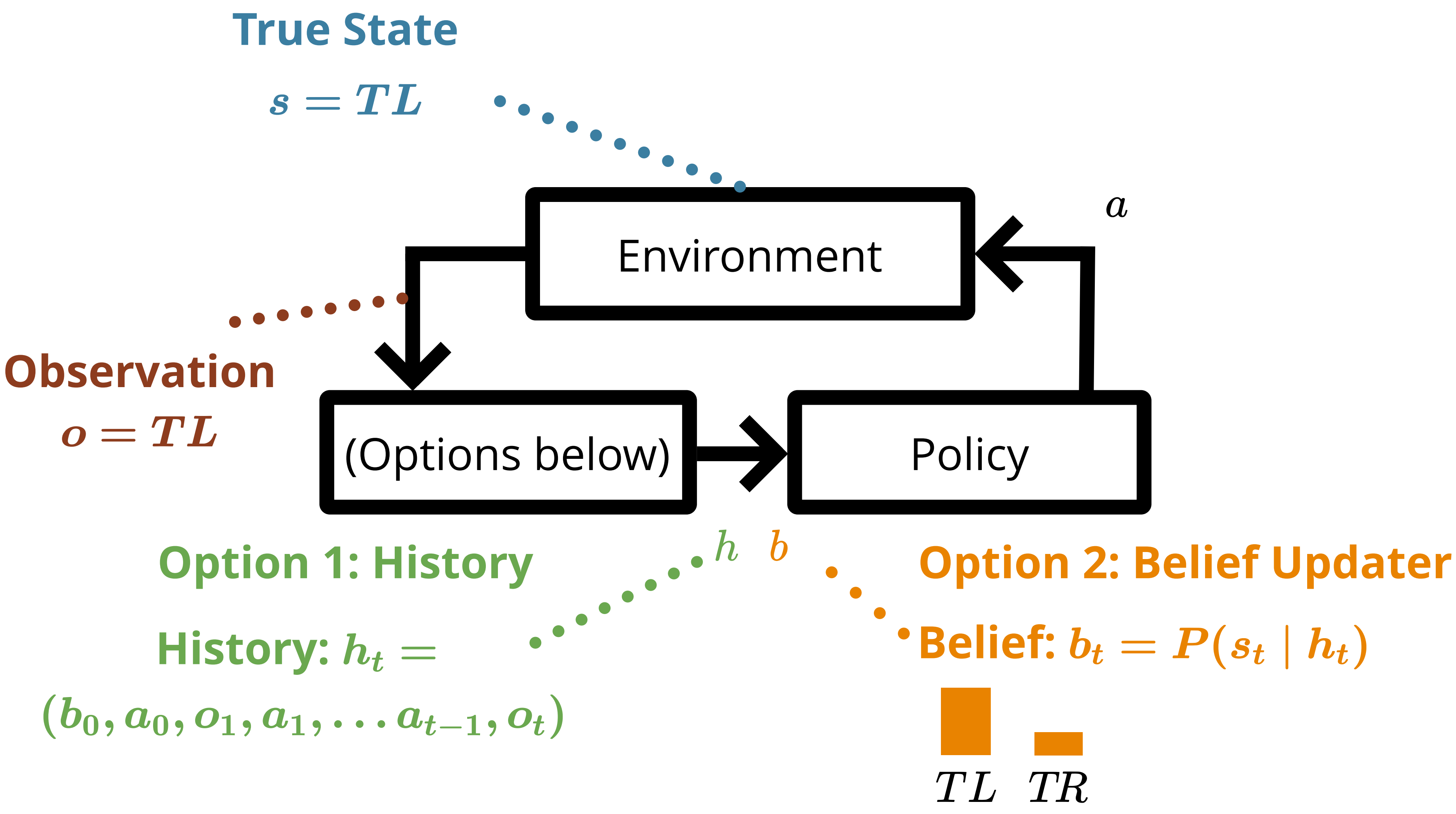

Exercise 1: Crying Baby Belief Update

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Starting at a $b(h) = 0$, calculate b' with $a = \neg f$ and $o = c$.

Guiding Quesiton

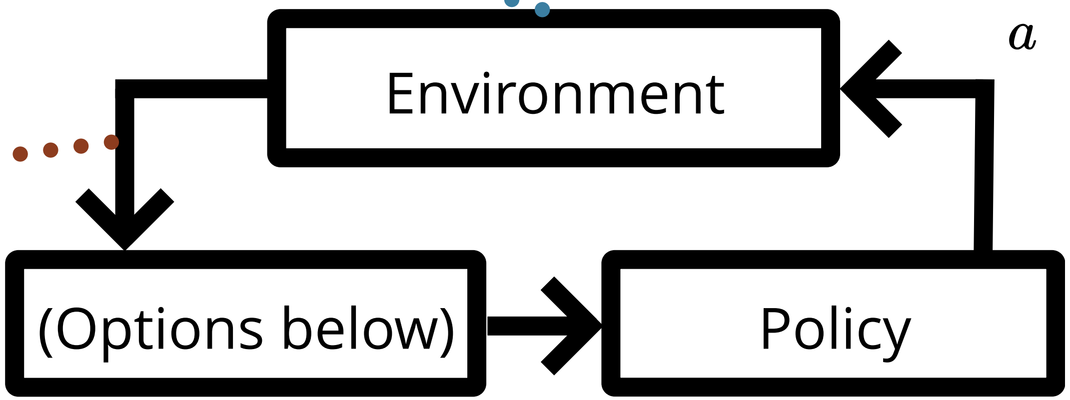
How do we calculate the optimal action in a POMDP?

True State

$s = TL$

Observation

$o = TL$



h b

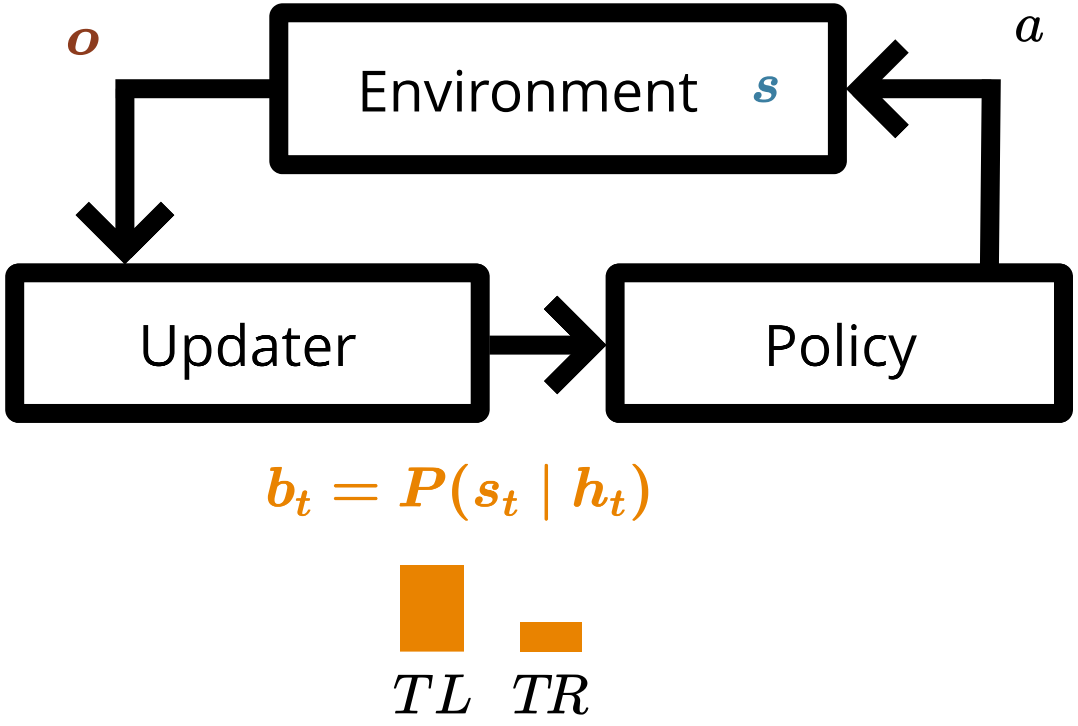
Option 1: History

History: $h_t = (b_0, a_0, o_1, a_1, \ldots a_{t-1}, o_t)$

Option 2: Belief Updater

Belief: $b_t = P(s_t \mid h_t)$

Belief-Space MDP



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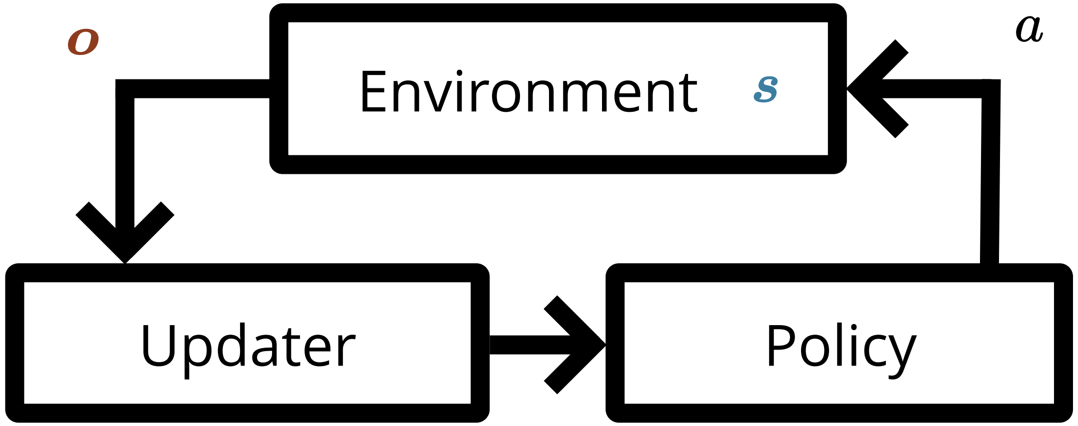

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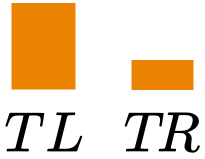

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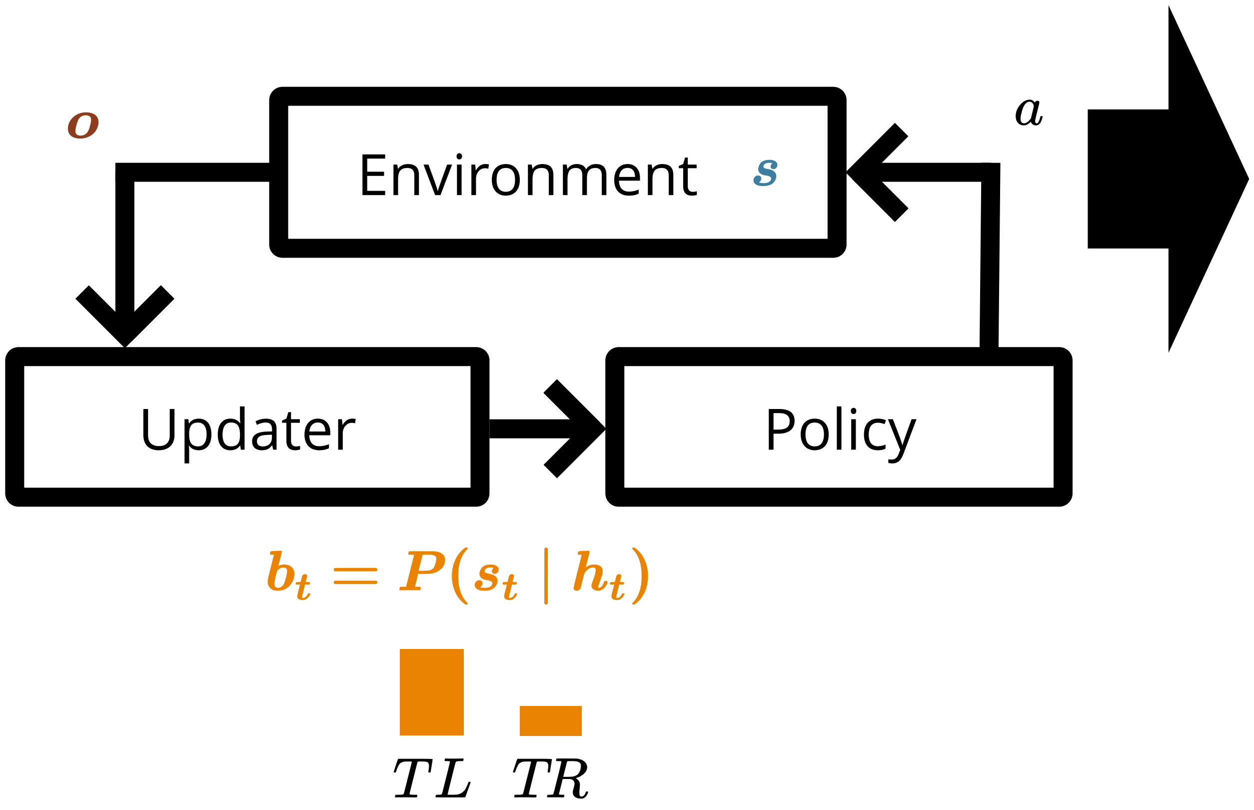

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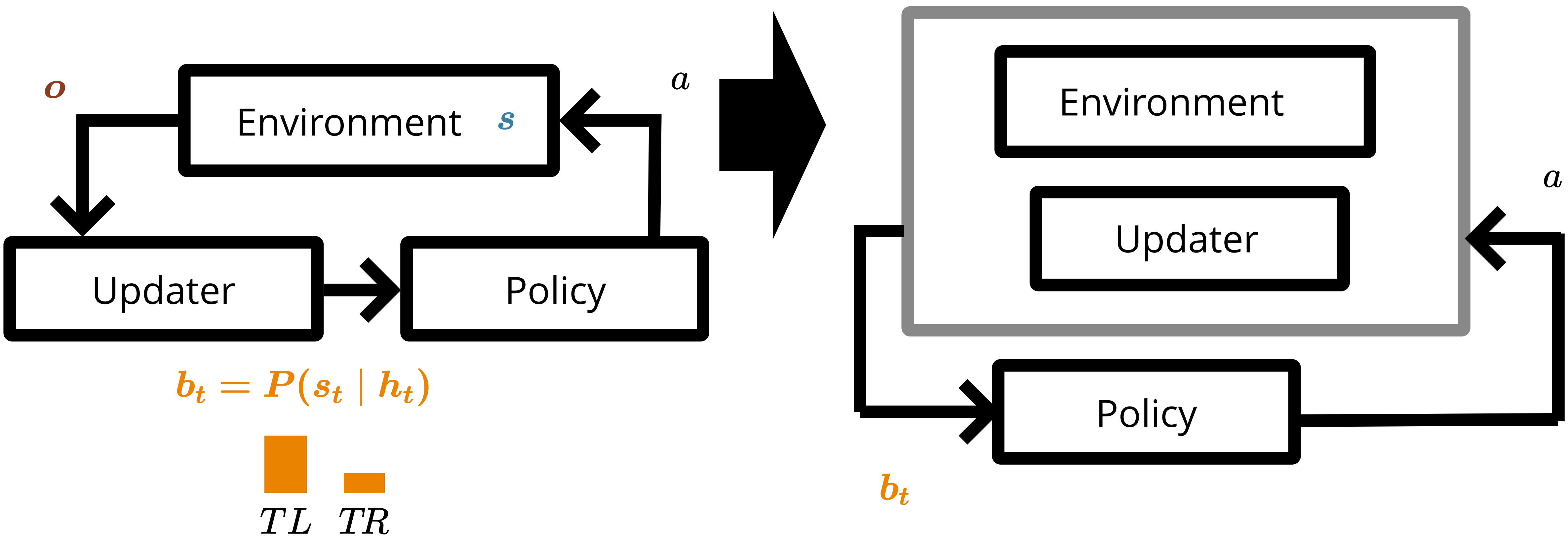

Belief Dynamics

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b(TL)

t

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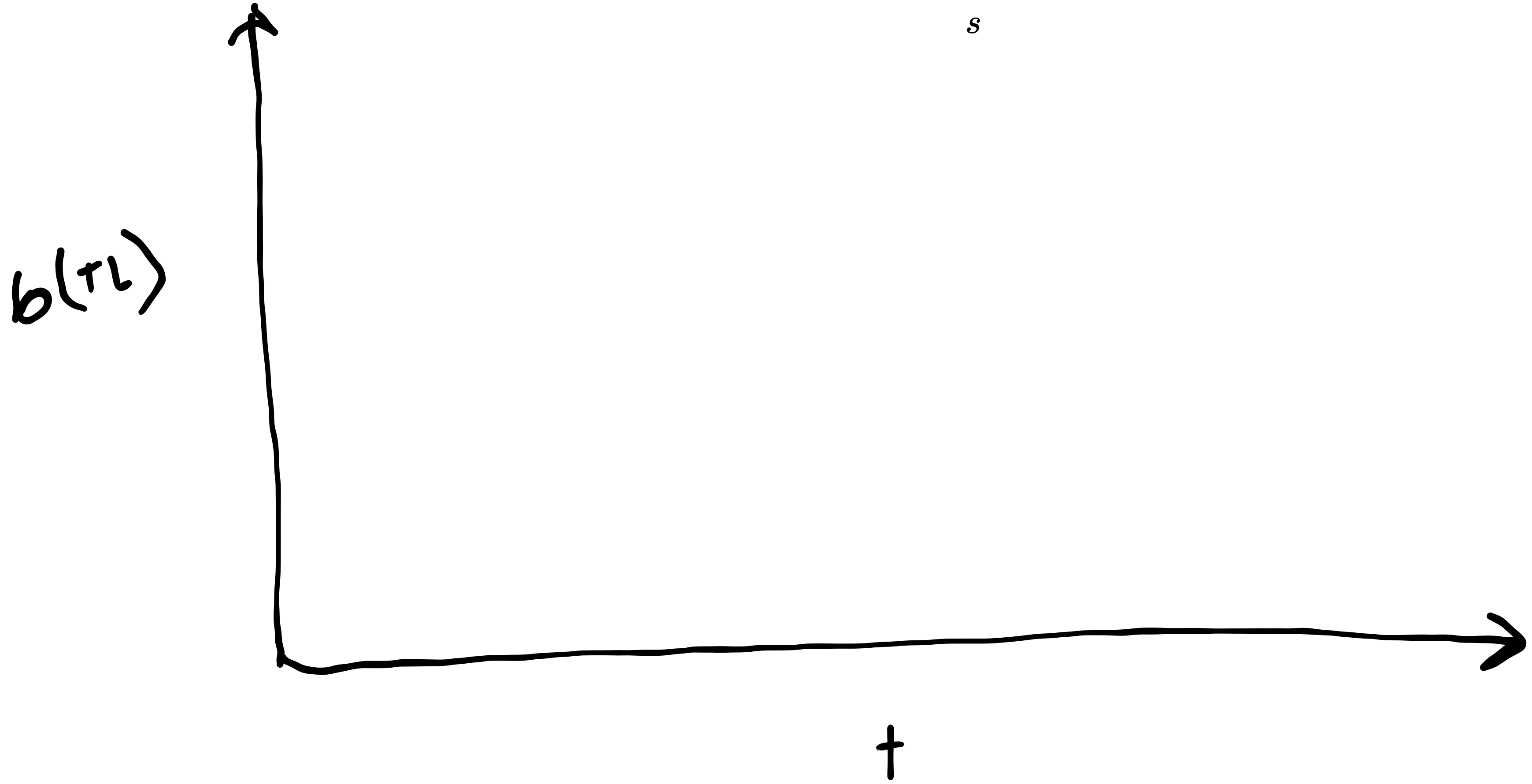

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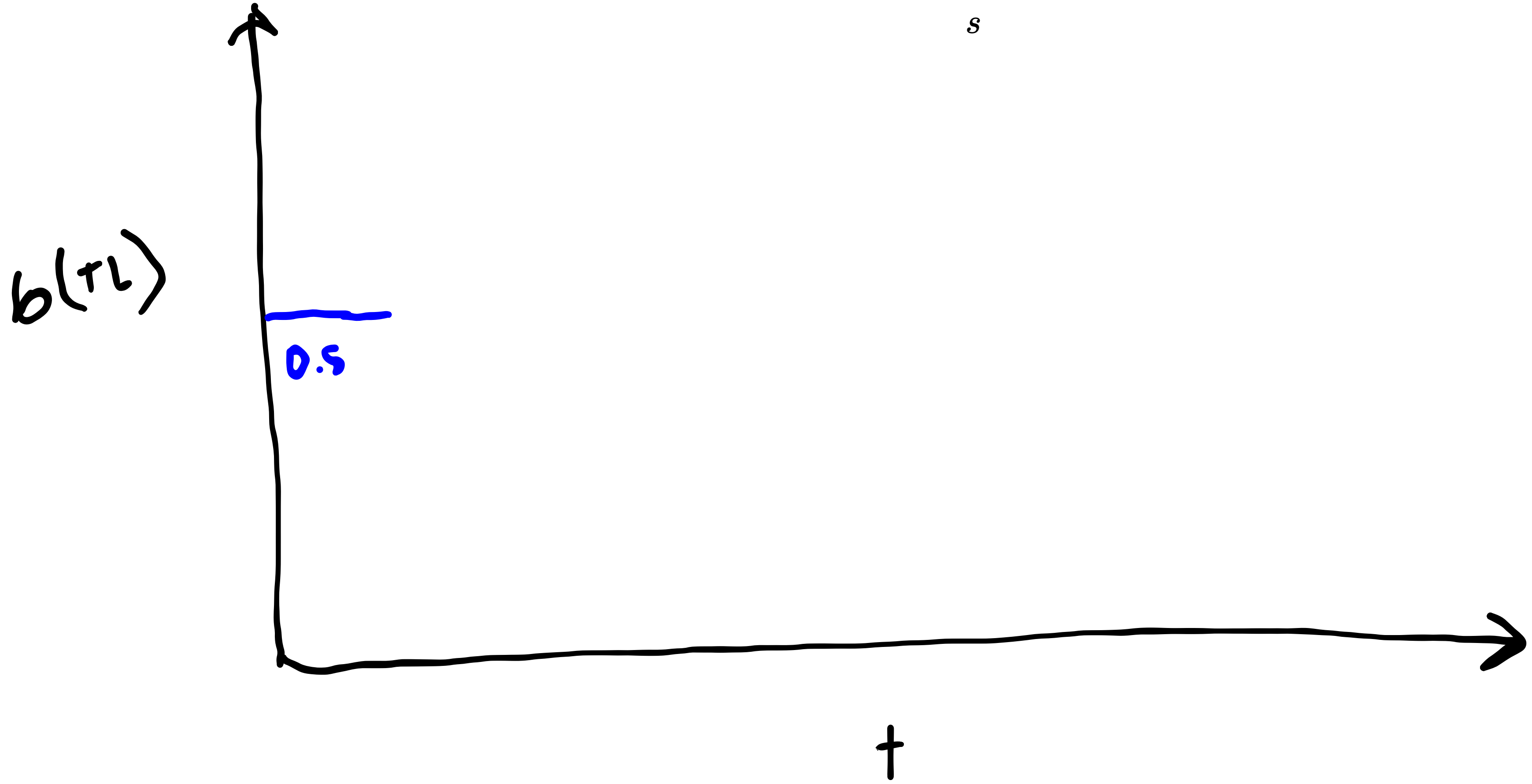

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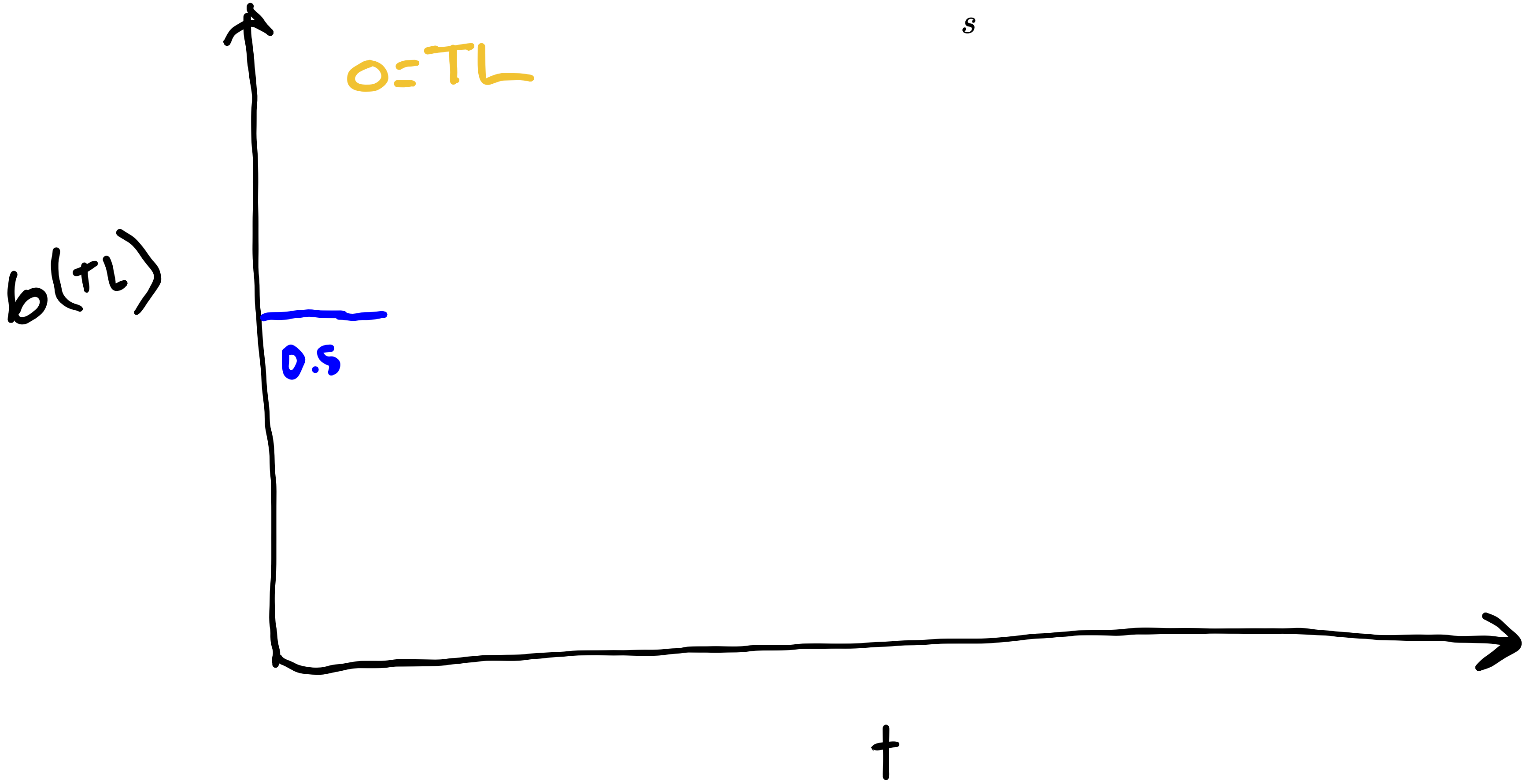

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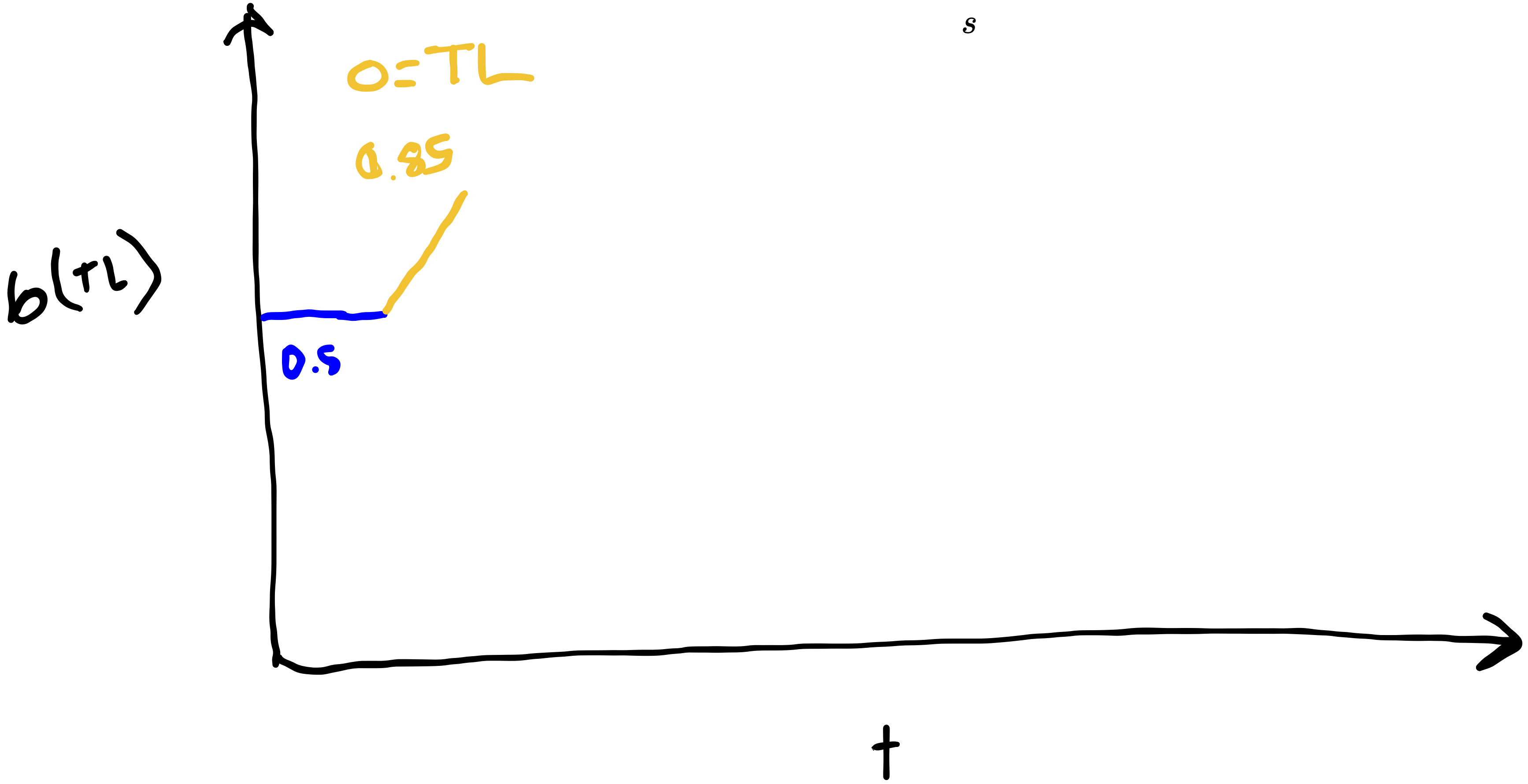

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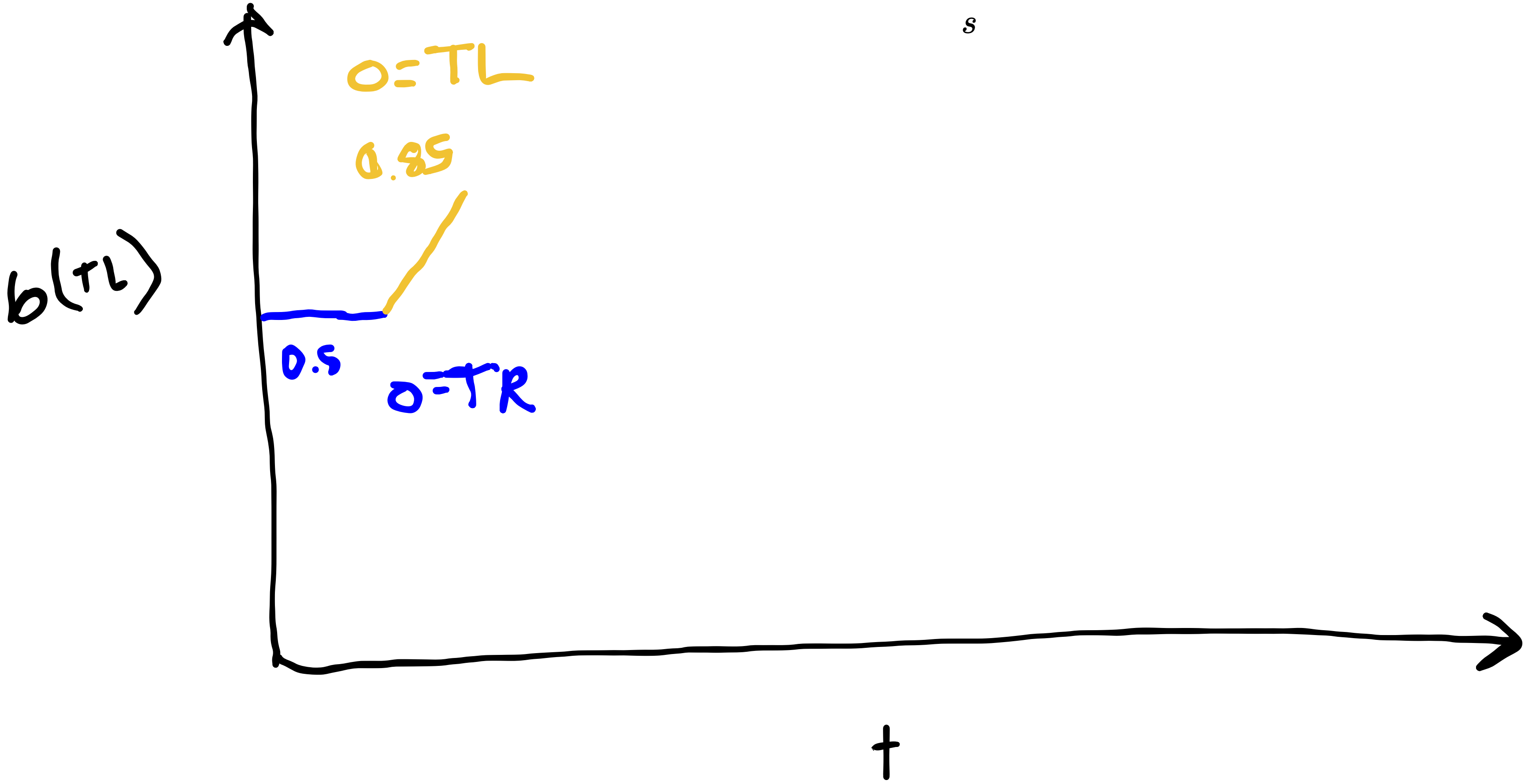

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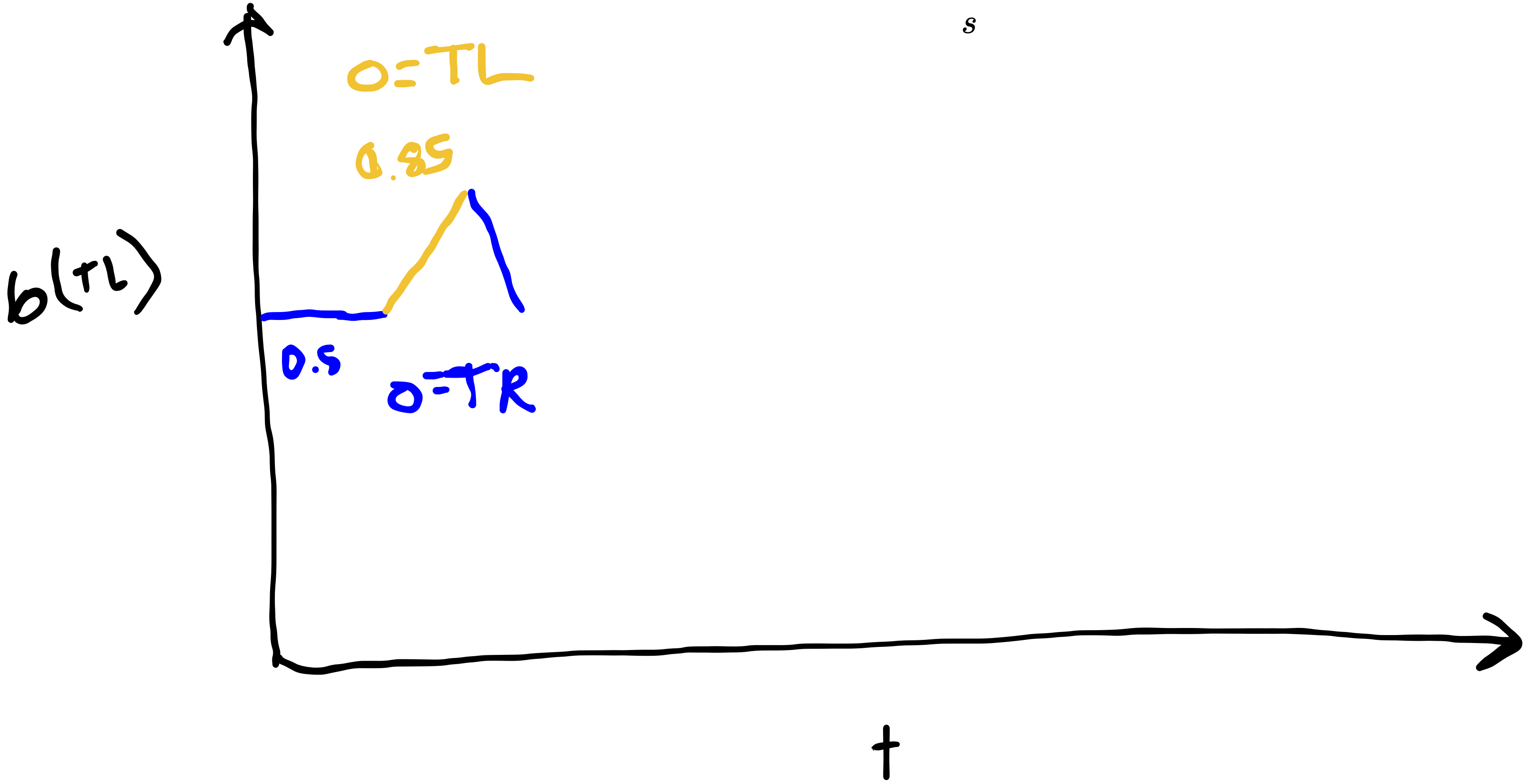

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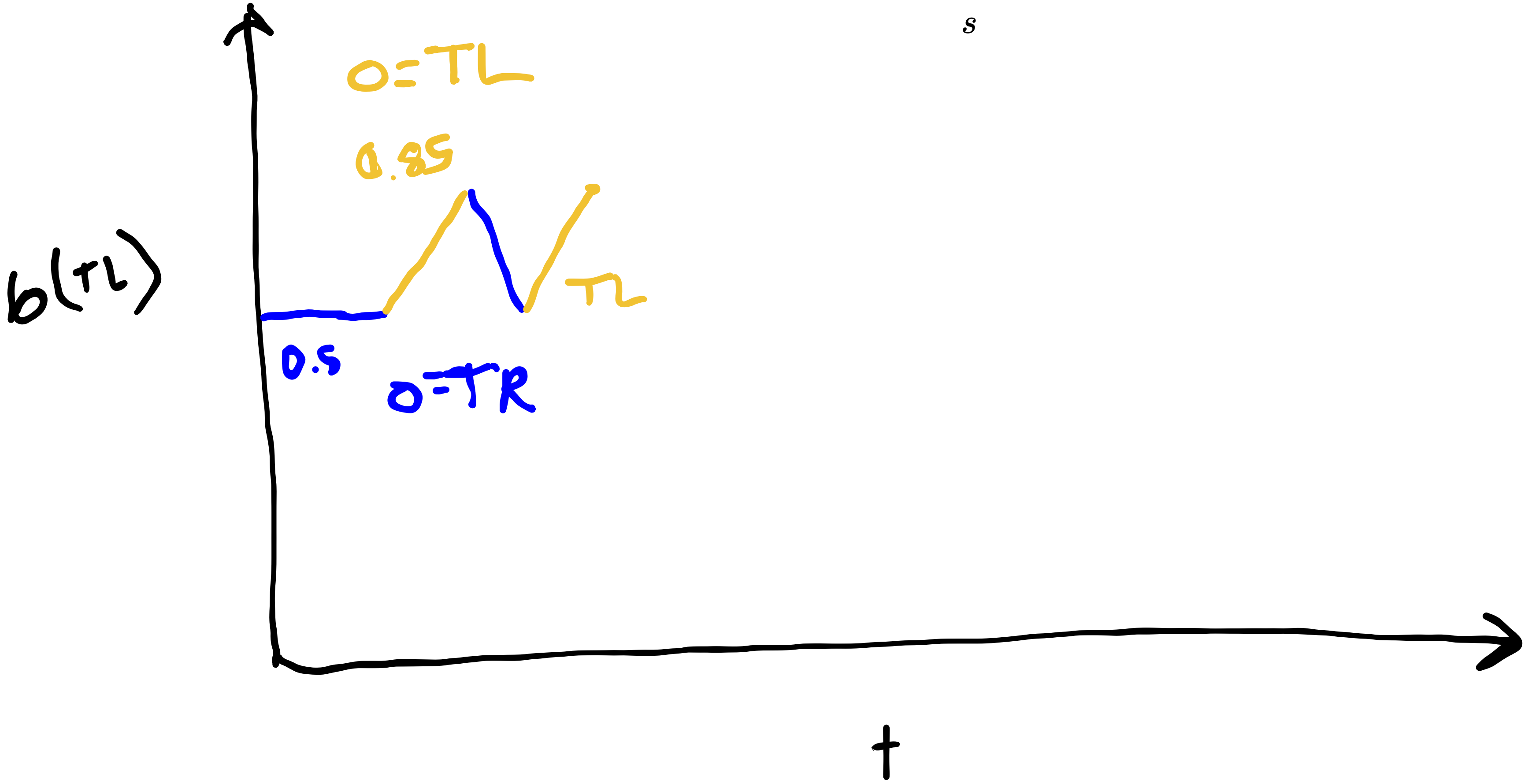

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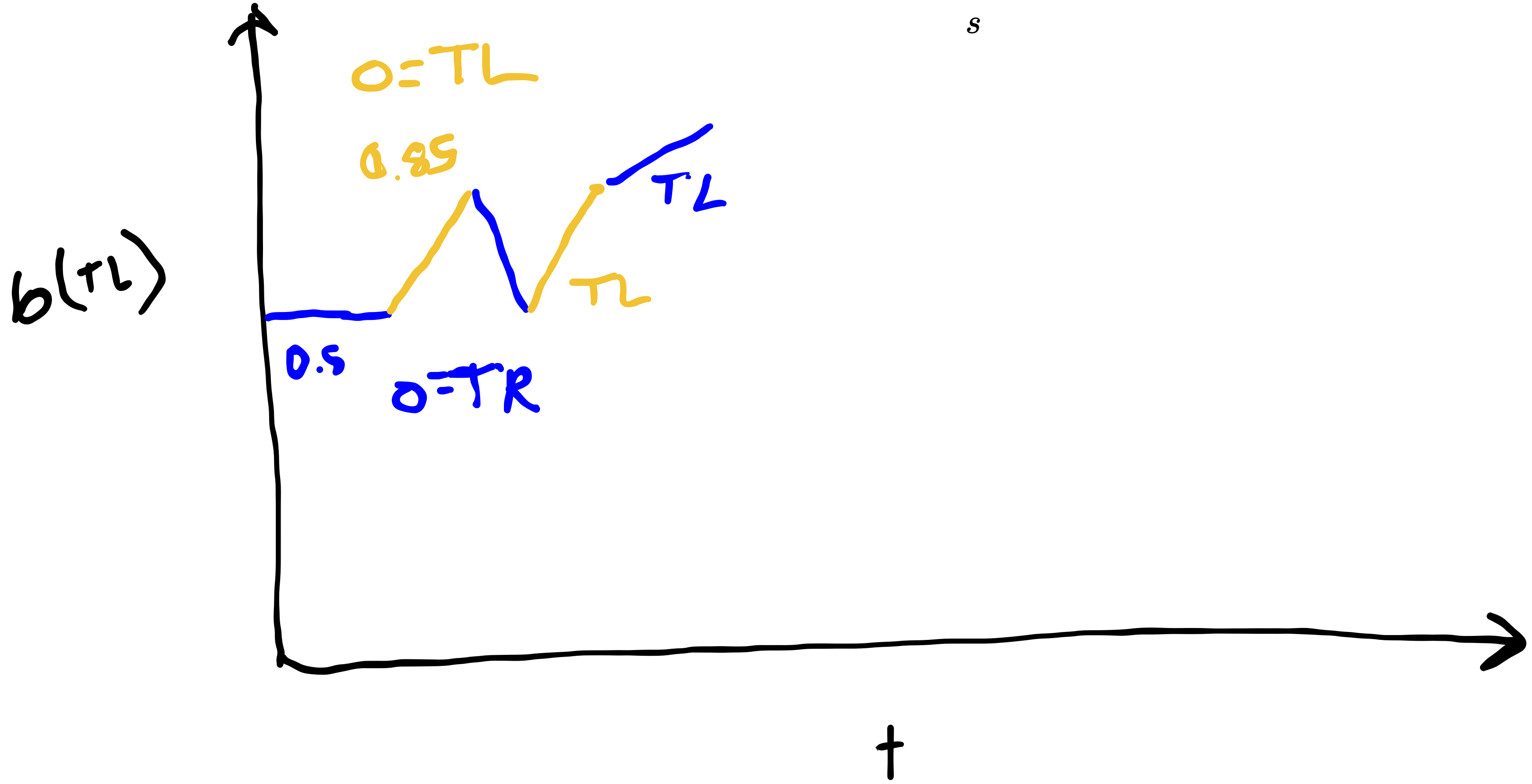

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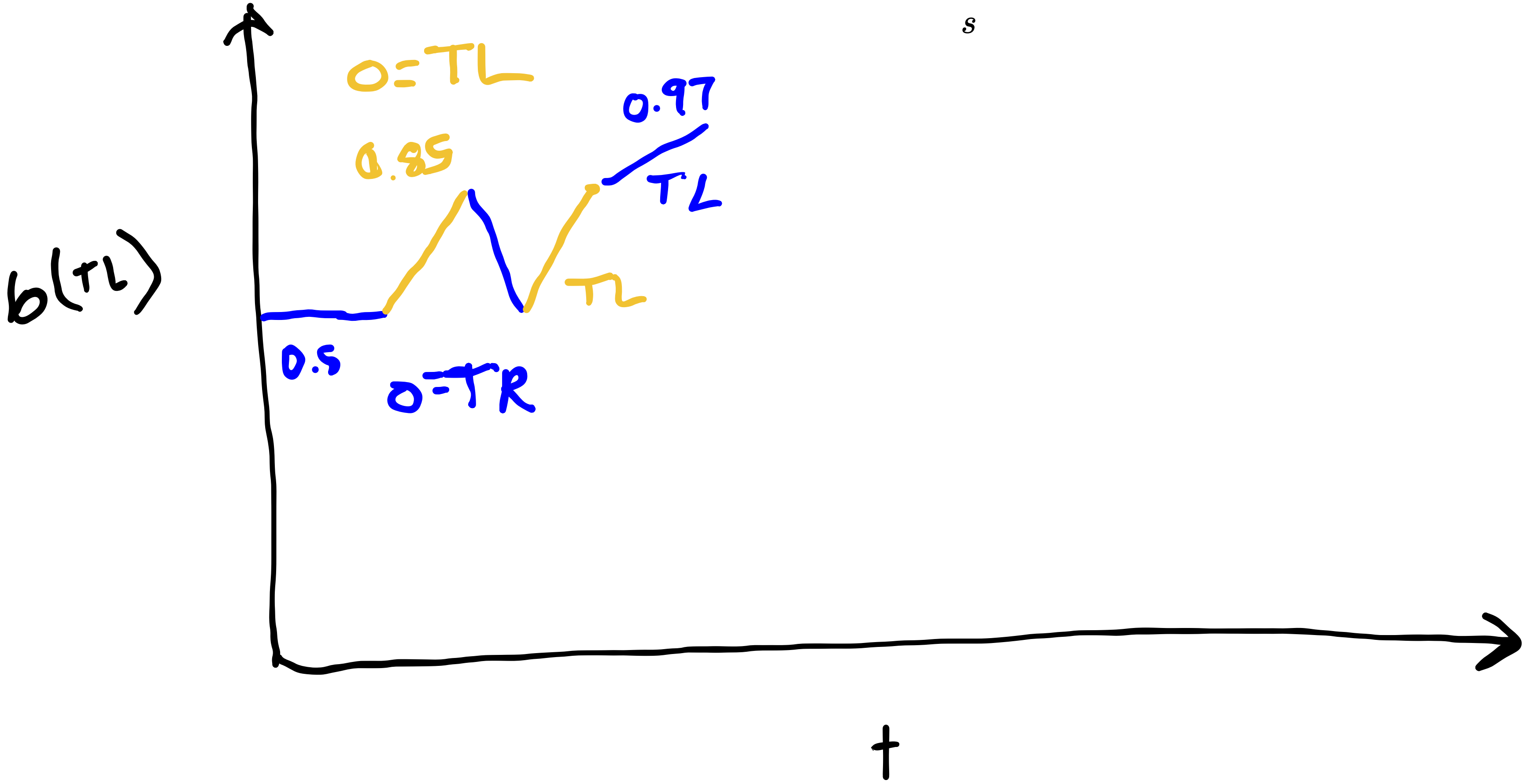

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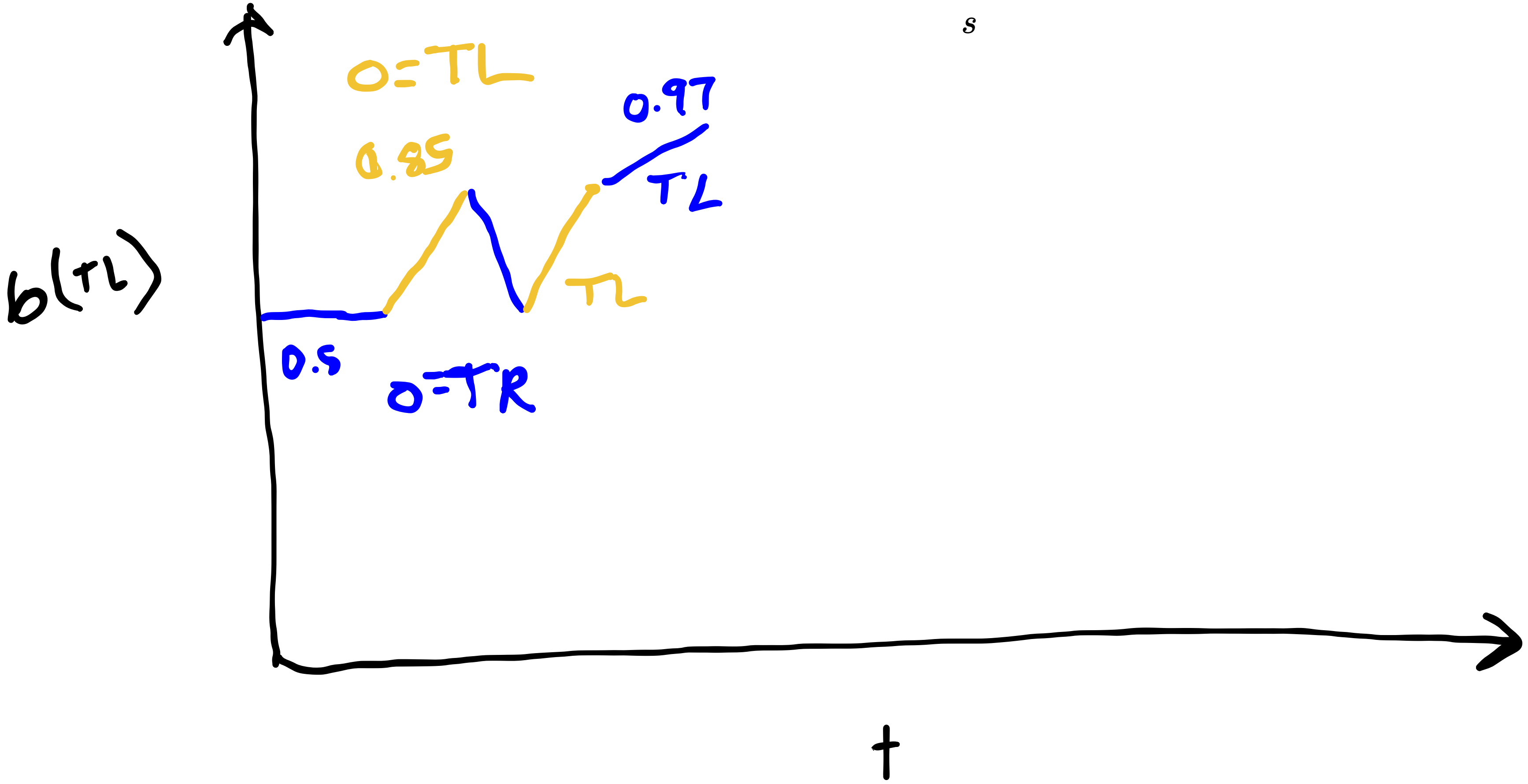

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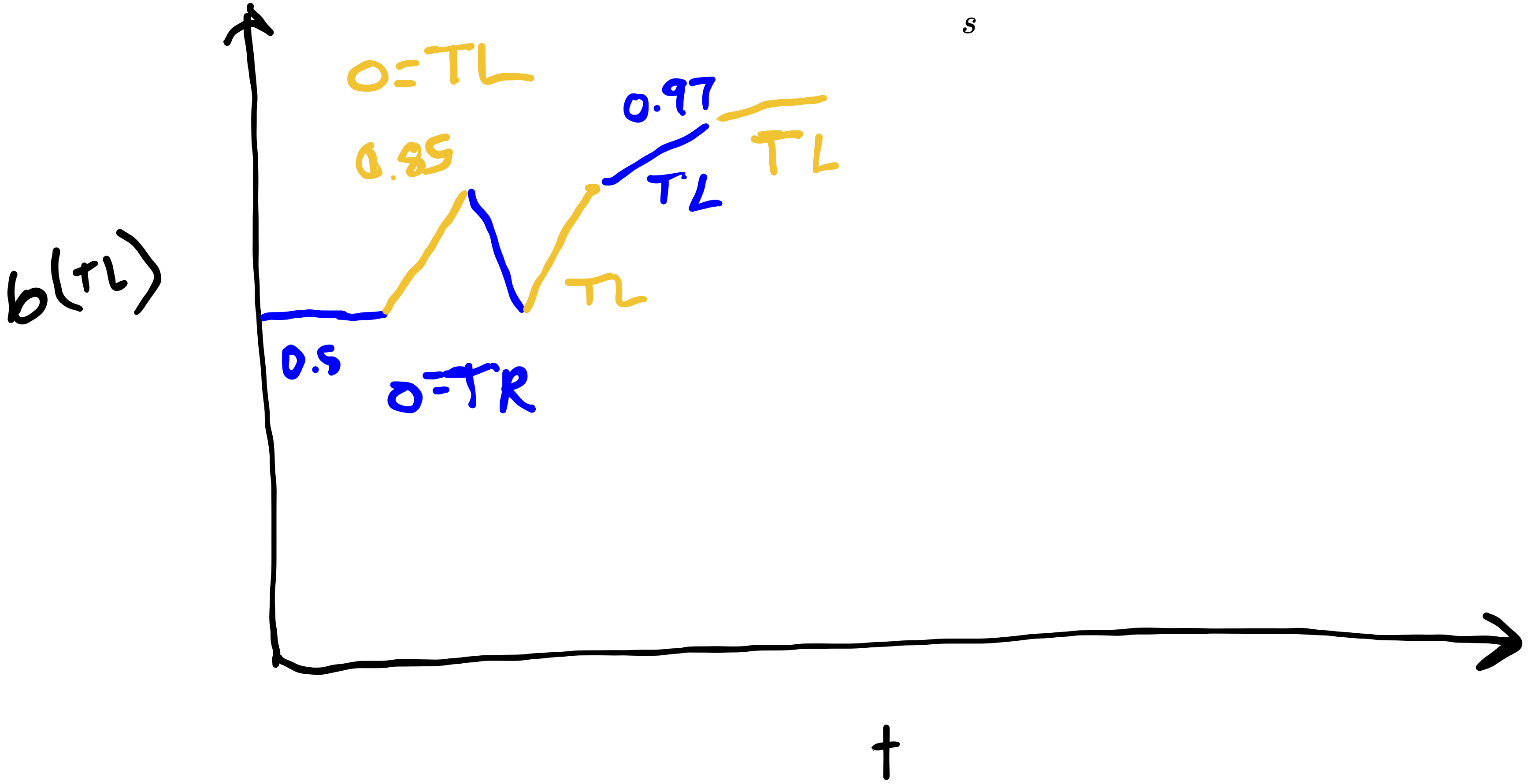


One-step utility

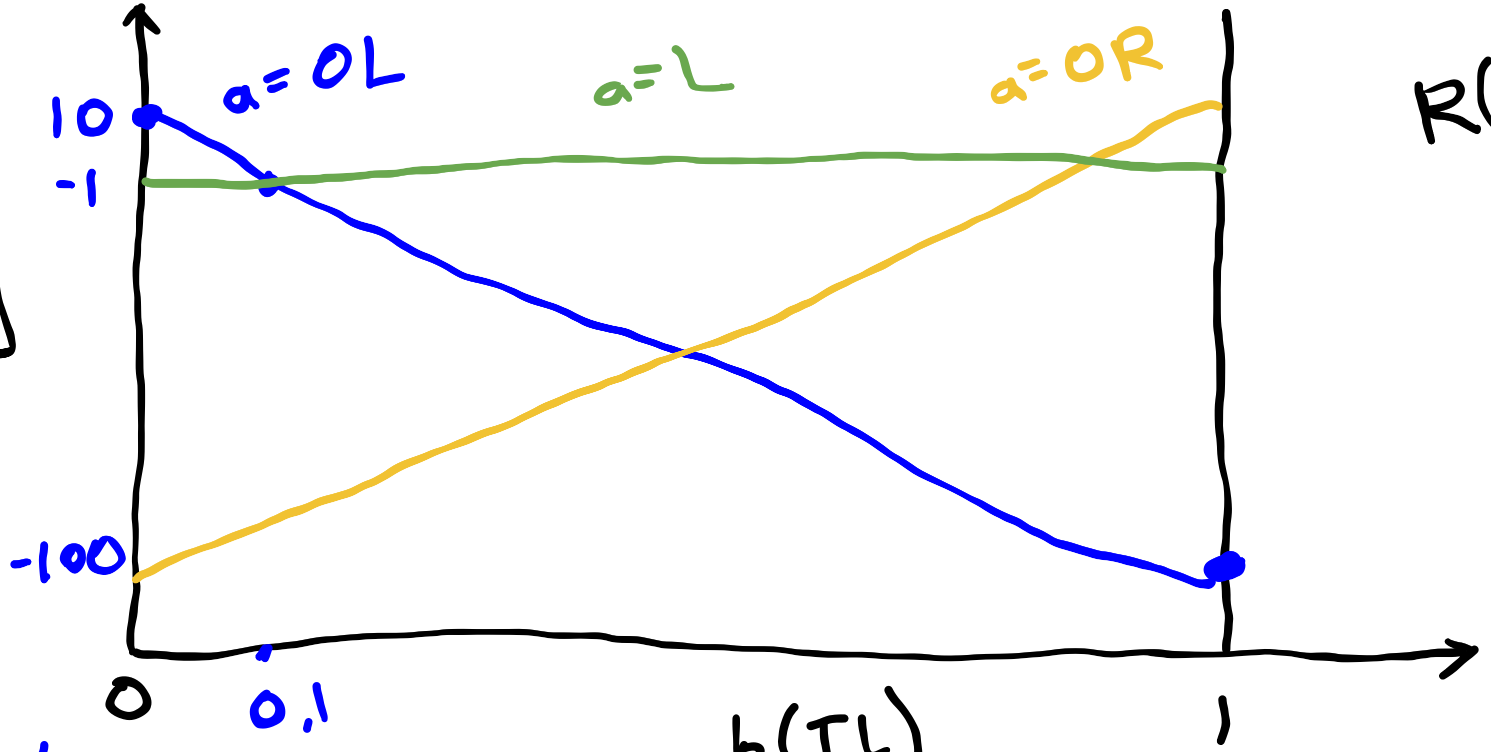
One-step utility

Reward: +10 empty door

−1 Listen

−100 Tiger

$$R(b,a) = \mathop{E}_{s\sim b}\left[R(s,a)\right]$$



$$R(b,a) = \bar{r}_a \cdot b$$

α-vector

$$R(b,a) = 10\cdot(1-b(TL)) - 100\, b(TL) = -110\, b + 10$$

Exercise 2: Crying Baby 1-Step Utility

$$S = \{h, \neg h\} \qquad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \qquad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \qquad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 \text{ if } s = h \\ 0 \text{ otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 \text{ if } a = f \\ 0 \text{ otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8)$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$

Draw the 1-step utility α-vectors for the Crying Baby problem.

Alpha Vectors for Conditional Plans

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Conditional Plans: fixed-depth history-based policies

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1 Step:

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1 Step: L OL OR

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Alpha Vectors for Conditional Plans

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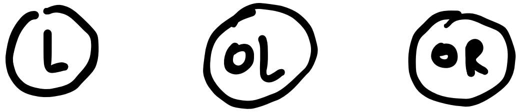

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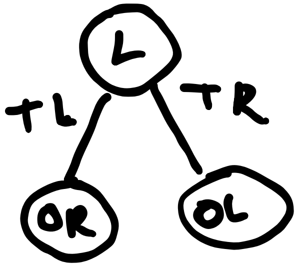

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Conditional Plans: fixed-depth history-based policies

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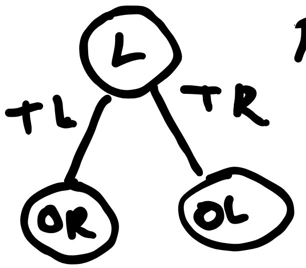

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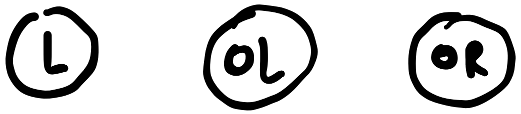

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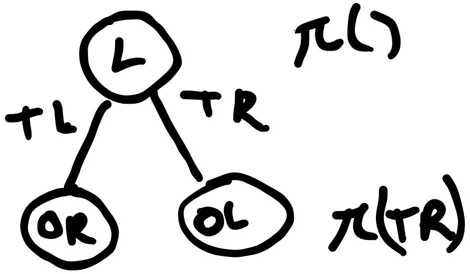

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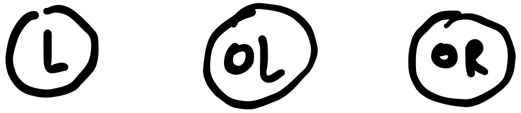

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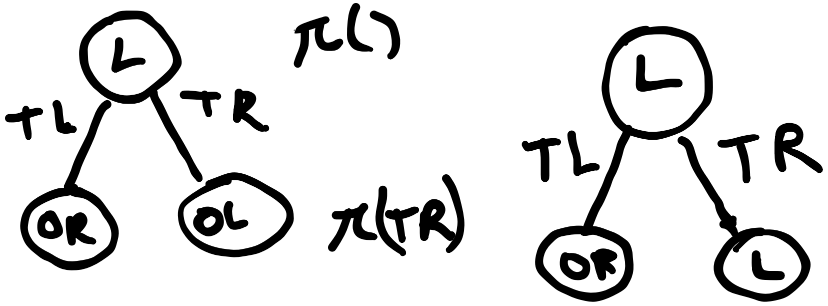

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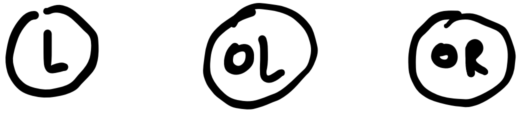
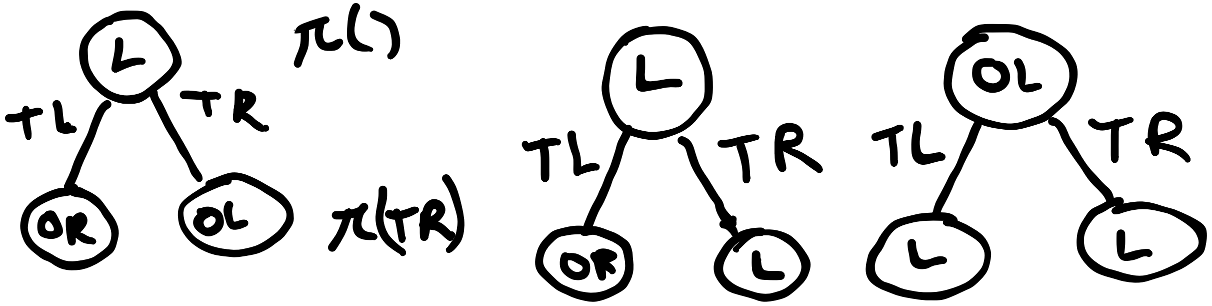
1 Step: 

2 Step: 

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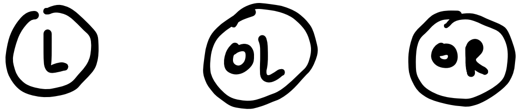

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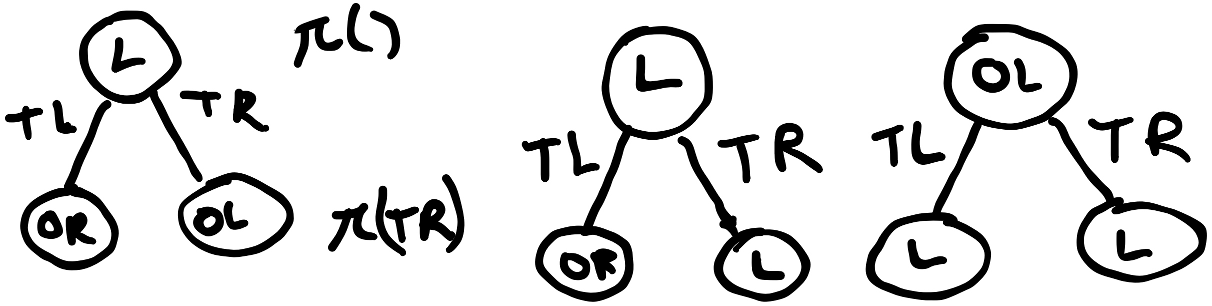

$$|A|^{\frac{(|O|^{h}-1)}{(|O|-1)}}$$

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

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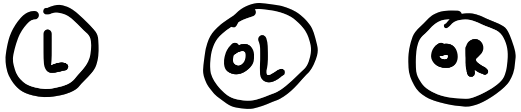

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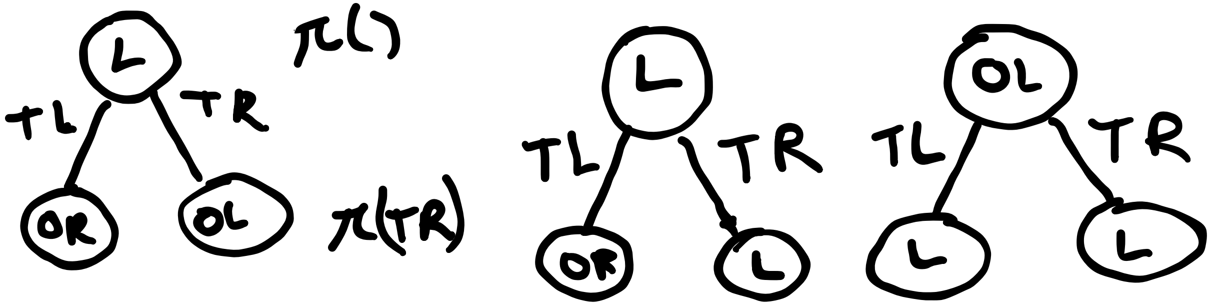

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27 two step plans!

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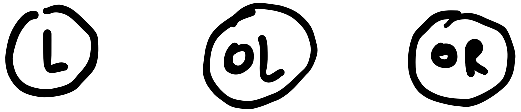

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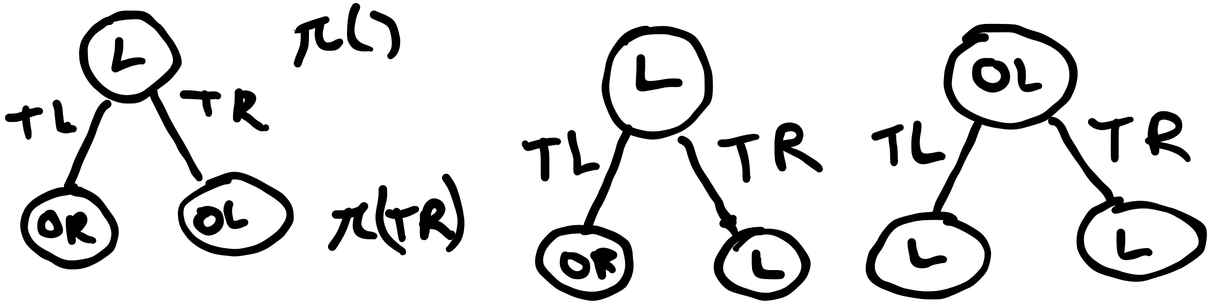

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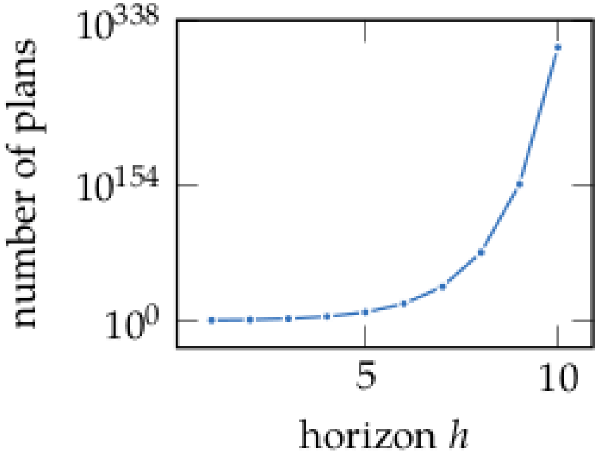

Alpha Vectors for Conditional Plans

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For 1-step: $U^\pi(s) = R(s, \pi())$

Alpha Vectors for Conditional Plans

For 1-step: $U^\pi(s) = R(s, \pi())$

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_o O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

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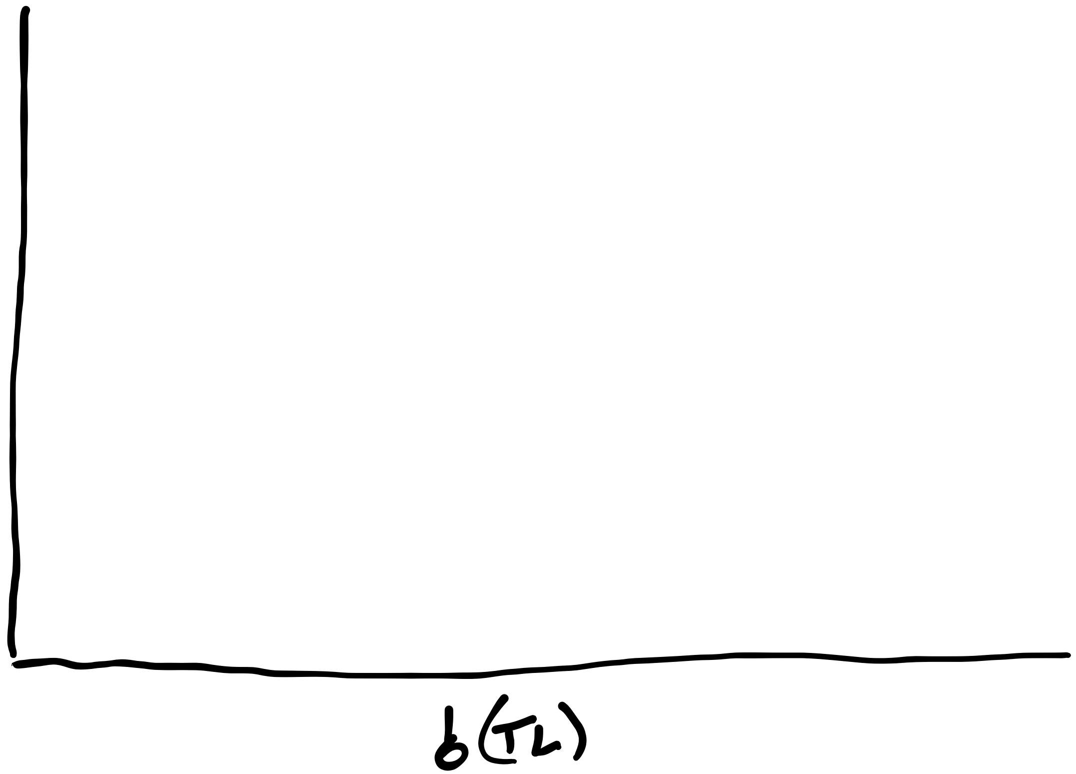

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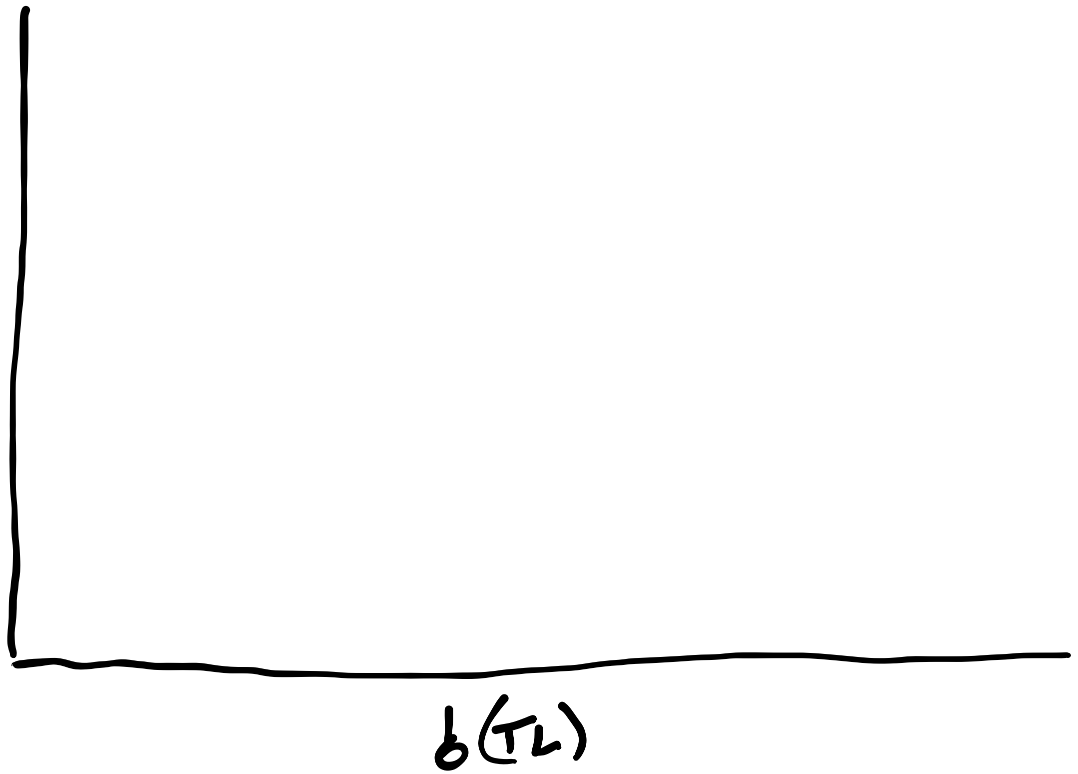

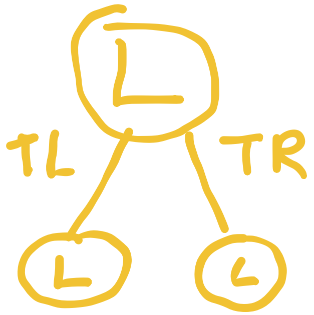

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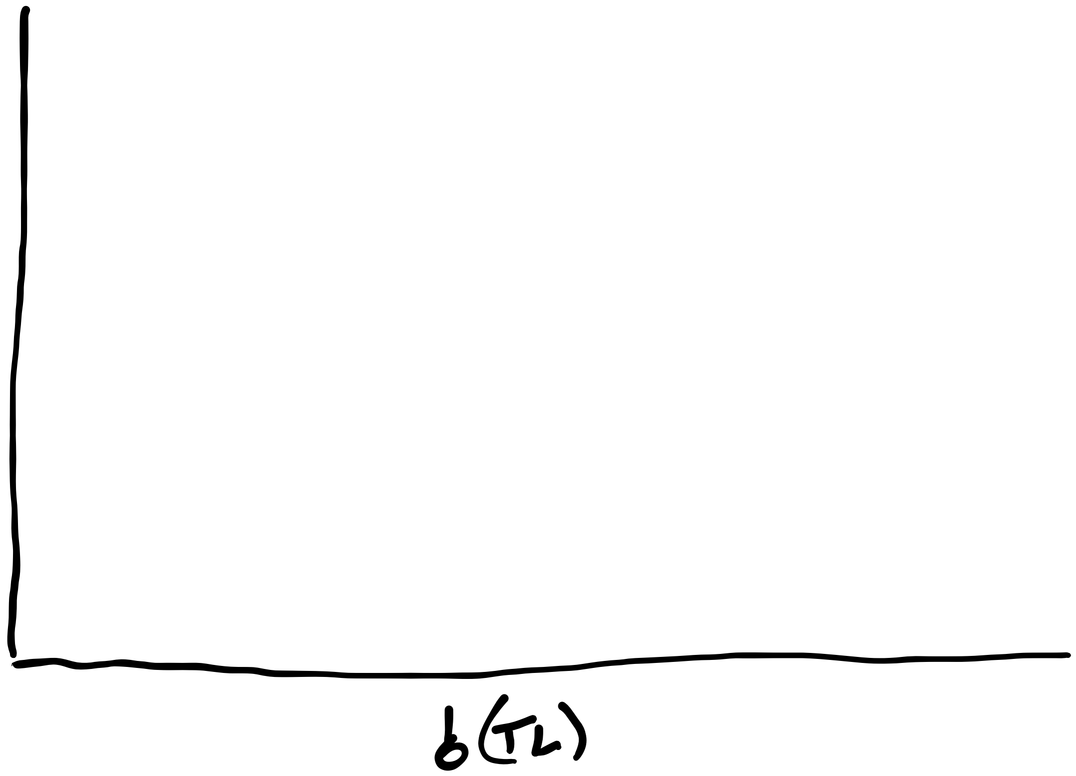

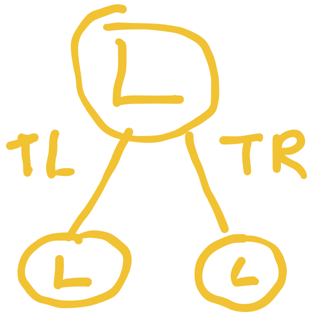

$$U^{\pi}(s) = -1 + \gamma(-1)$$

Alpha Vectors for Conditional Plans

$$U^{\pi}(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_{o} O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

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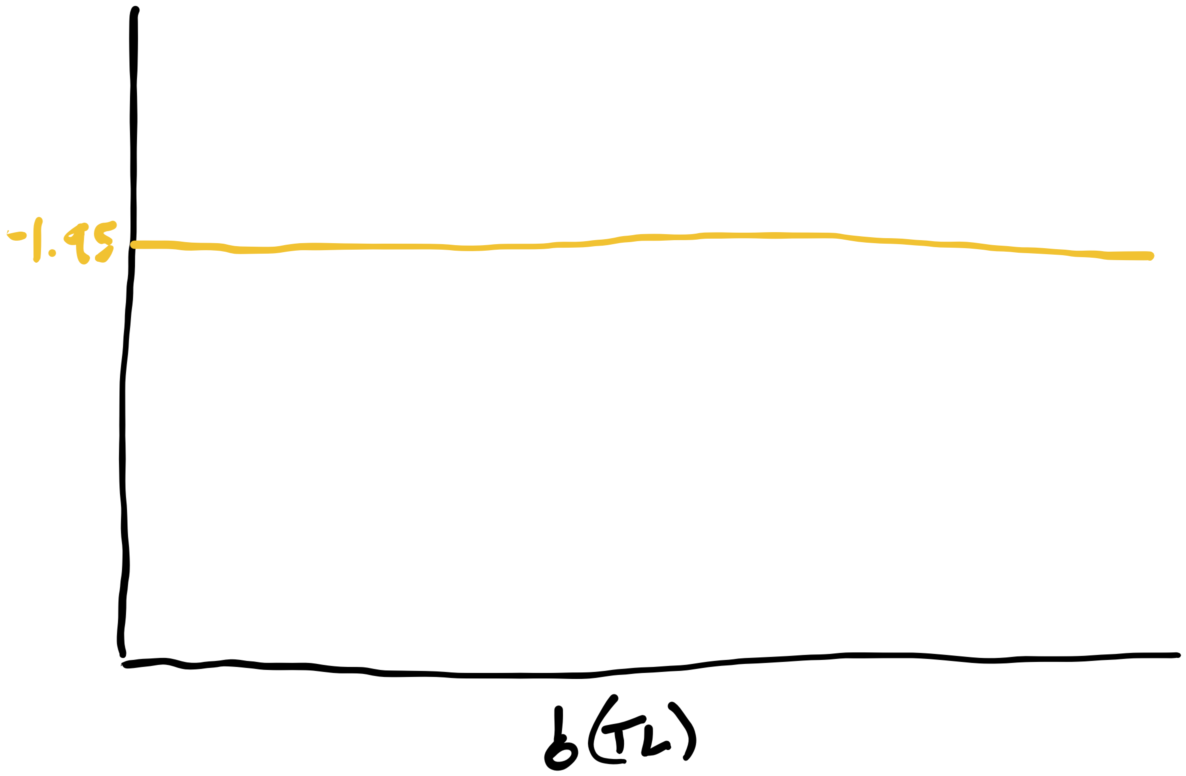

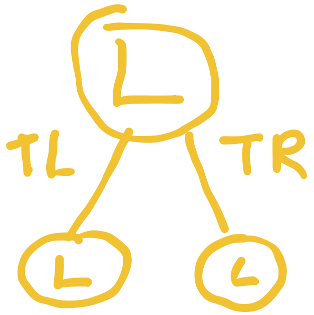

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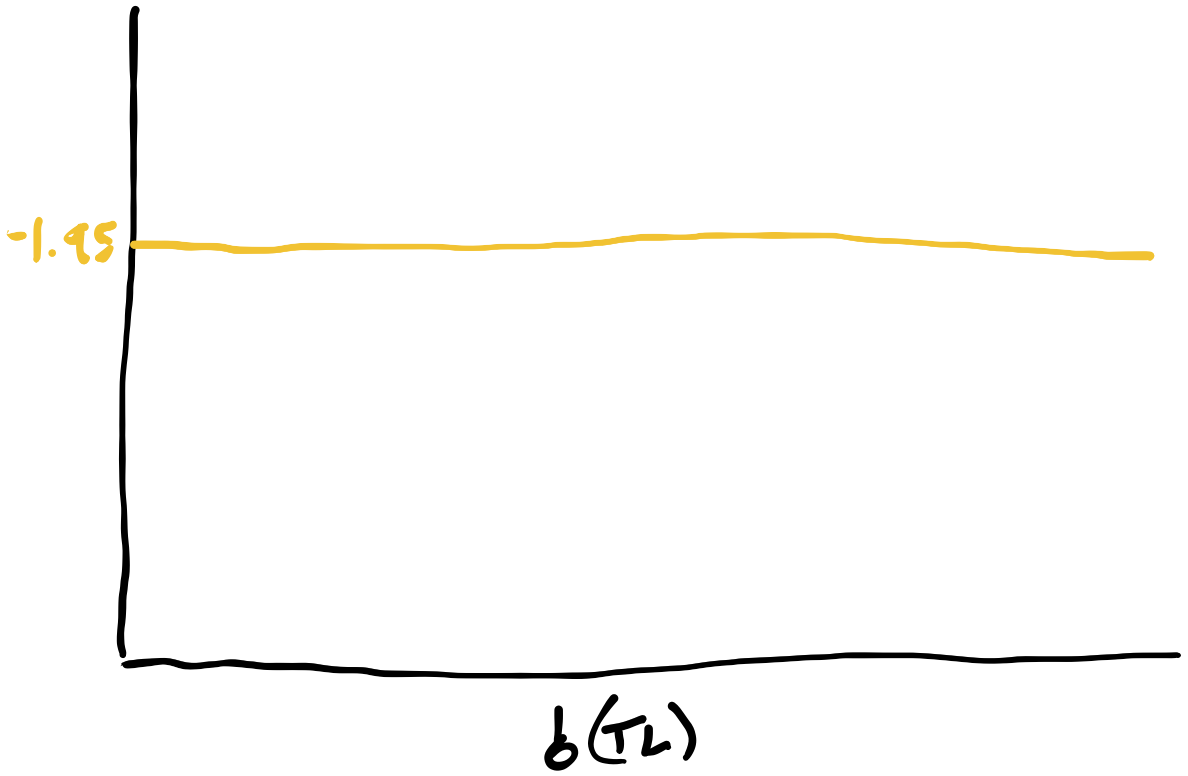

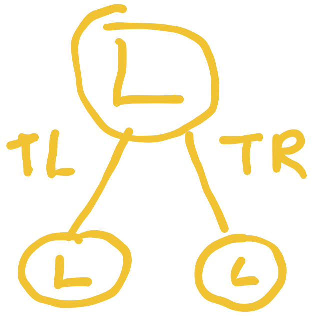

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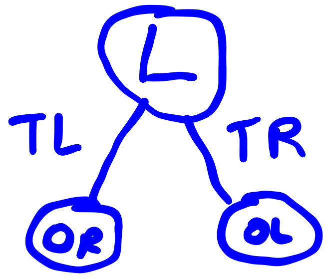

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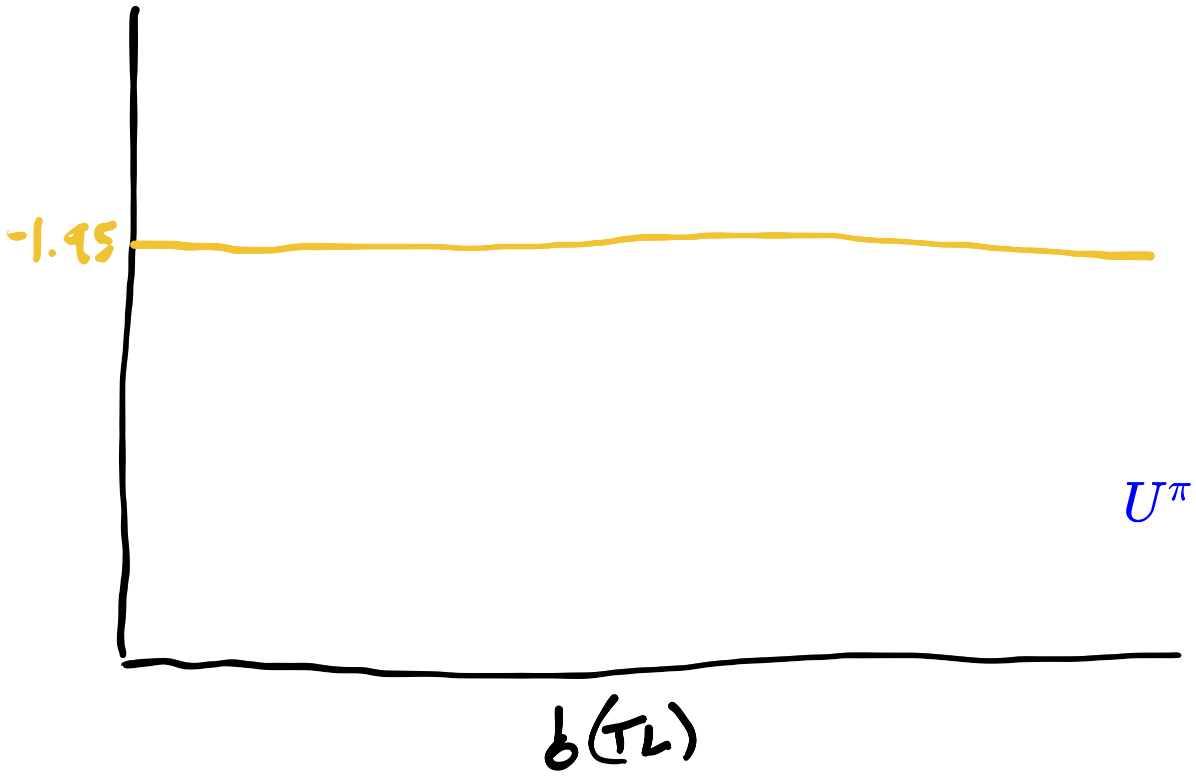
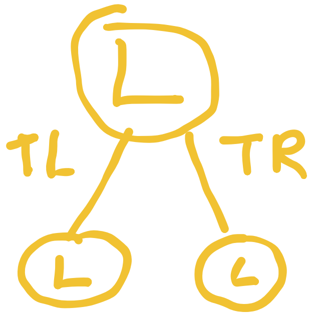
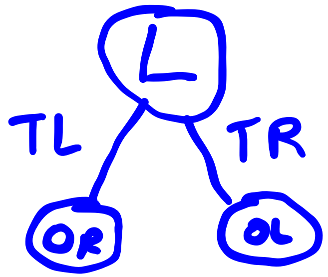






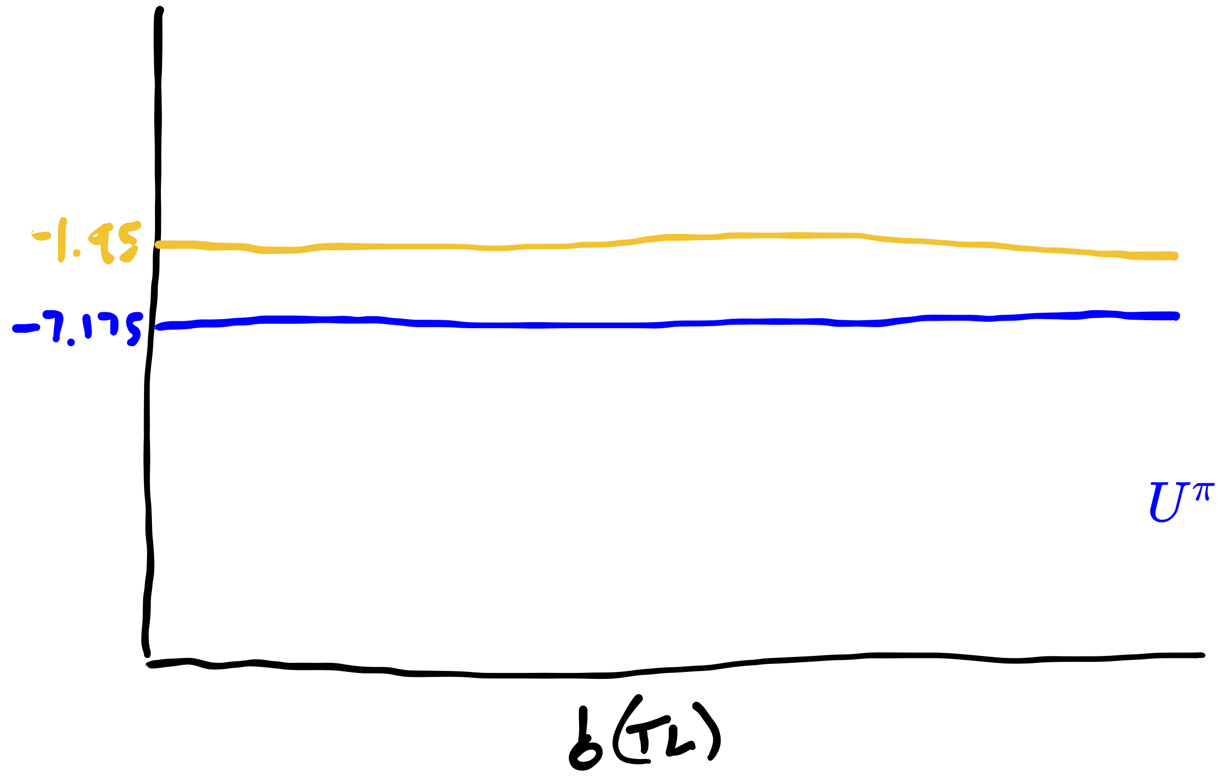
$$U^\pi(s) = -1 + \gamma(-1)$$

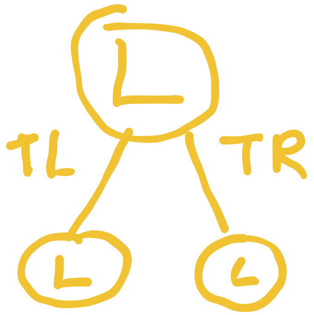
$$U^\pi(TL) = -1 + 0.95(1(0.85 \times 10 + 0.15 \times -100))$$

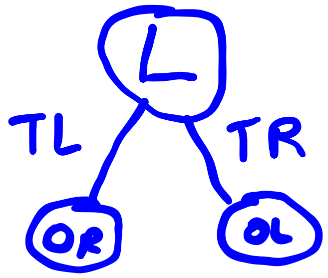
Alpha Vectors for Conditional Plans

$$U^{\pi}(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_{o} O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

For 1-step: $U^{\pi}(s) = R(s, \pi())$







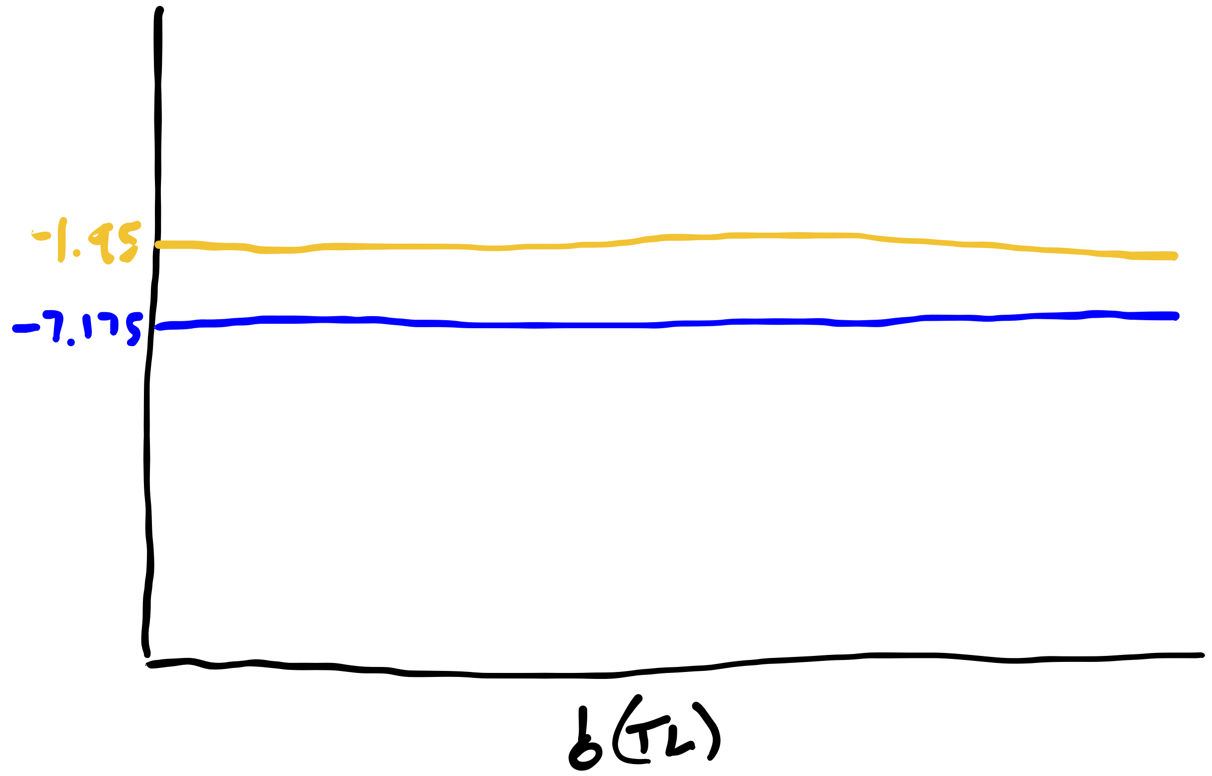
$$U^{\pi}(s) = -1 + \gamma(-1)$$

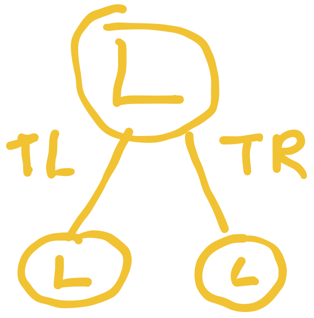
$$\begin{aligned} U^{\pi}(TL) &= -1 + 0.95(1(0.85 \times 10 + 0.15 \times -100)) \\ &= -7.175 \end{aligned}$$

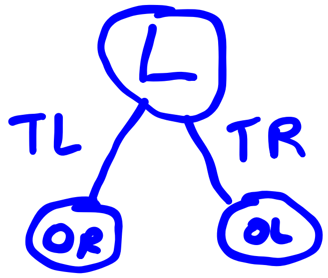
Alpha Vectors for Conditional Plans

$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_o O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

For 1-step: $U^\pi(s) = R(s, \pi())$







Alpha Vectors for Conditional Plans

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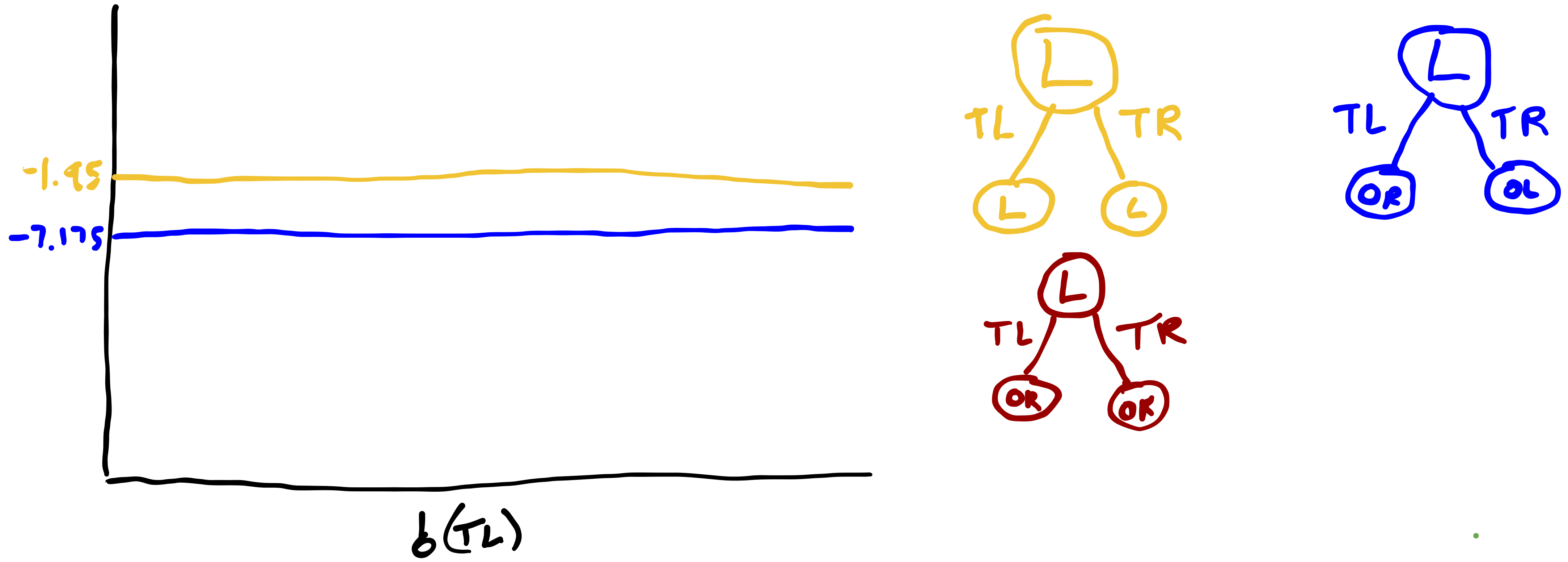

Alpha Vectors for Conditional Plans

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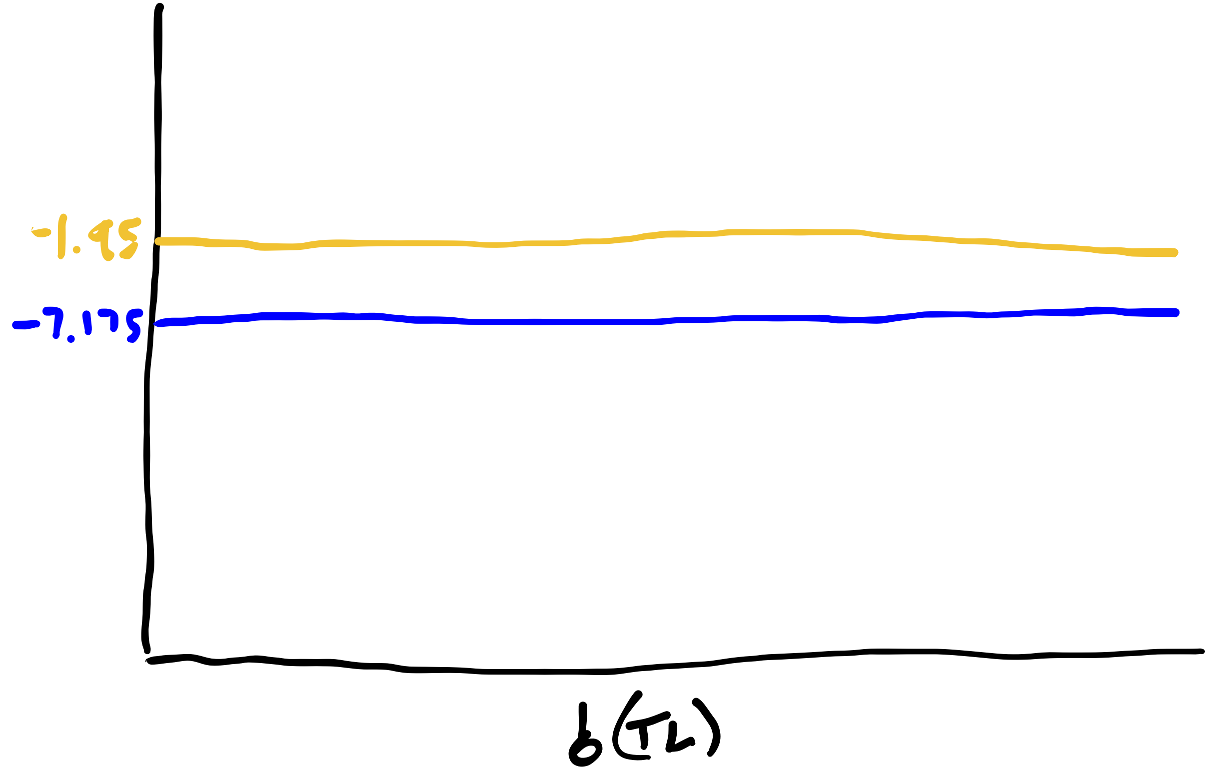

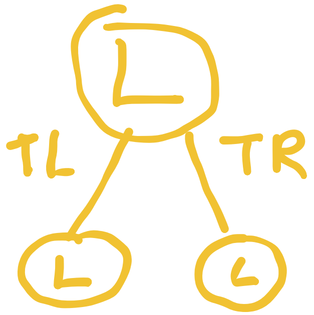

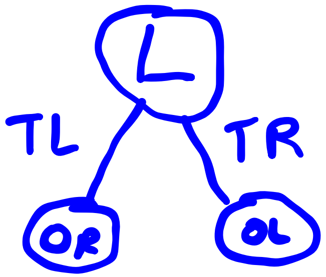

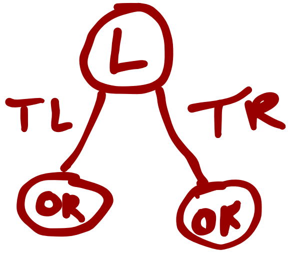

$U^\pi(TL) = -1 + \gamma\, 10$

Alpha Vectors for Conditional Plans

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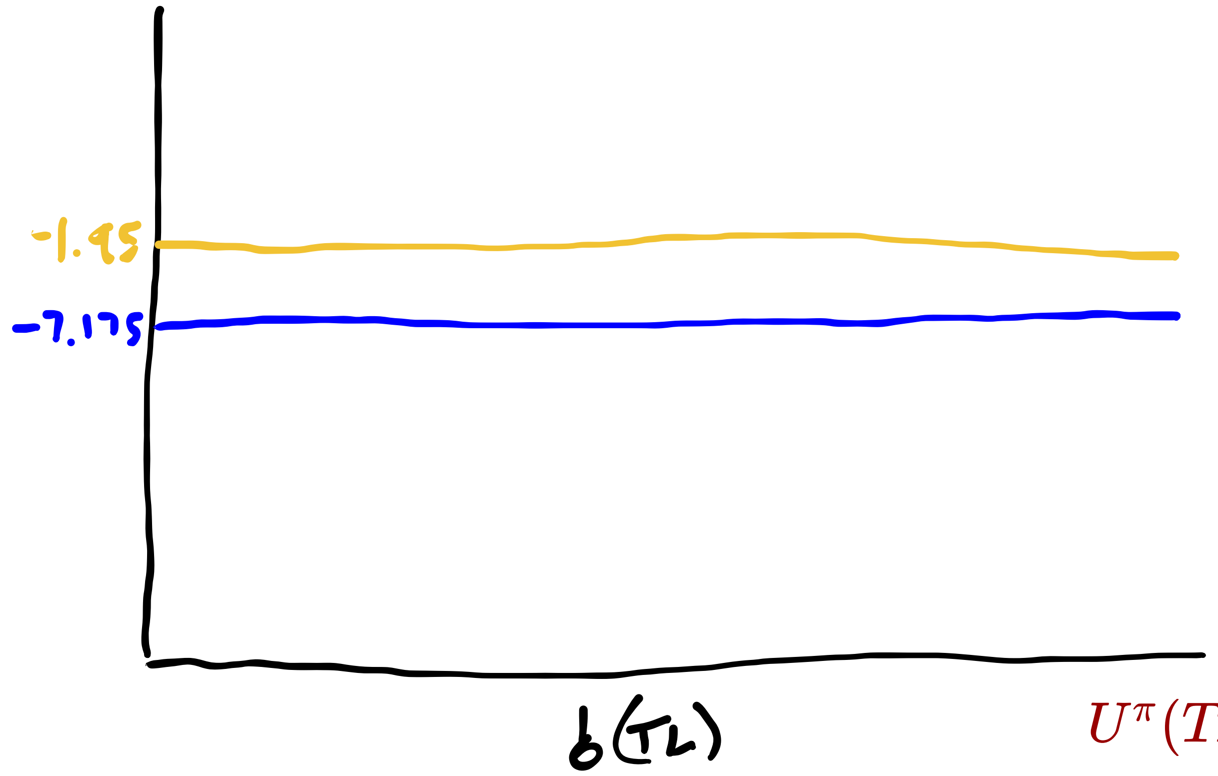

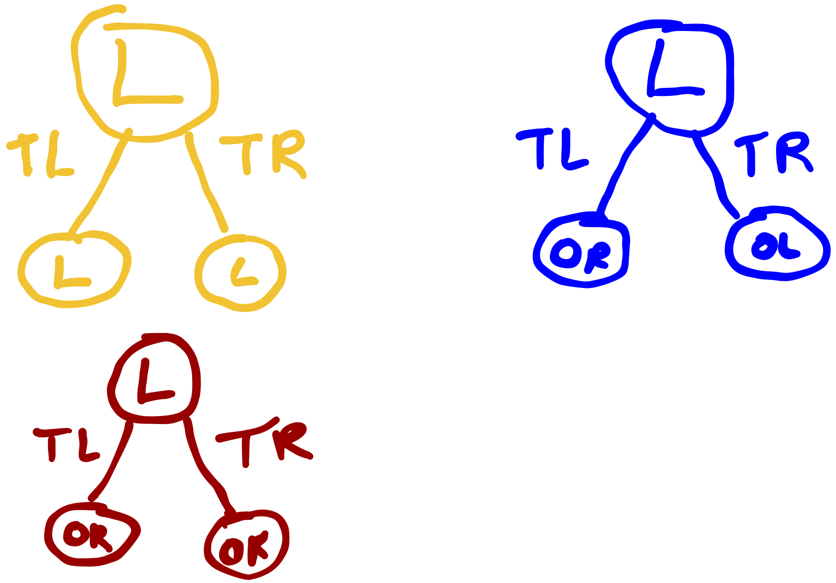

$U^\pi(TL) = -1 + \gamma\, 10$

$U^\pi(TR) = -1 + \gamma(-100)$

Alpha Vectors for Conditional Plans

$$U^\pi(s) = R(s,\pi()) + \gamma\left[\sum_{s'} T(s' \mid s,\pi()) \sum_o O(o \mid \pi(), s') U^{\pi(o)}(s')\right]$$

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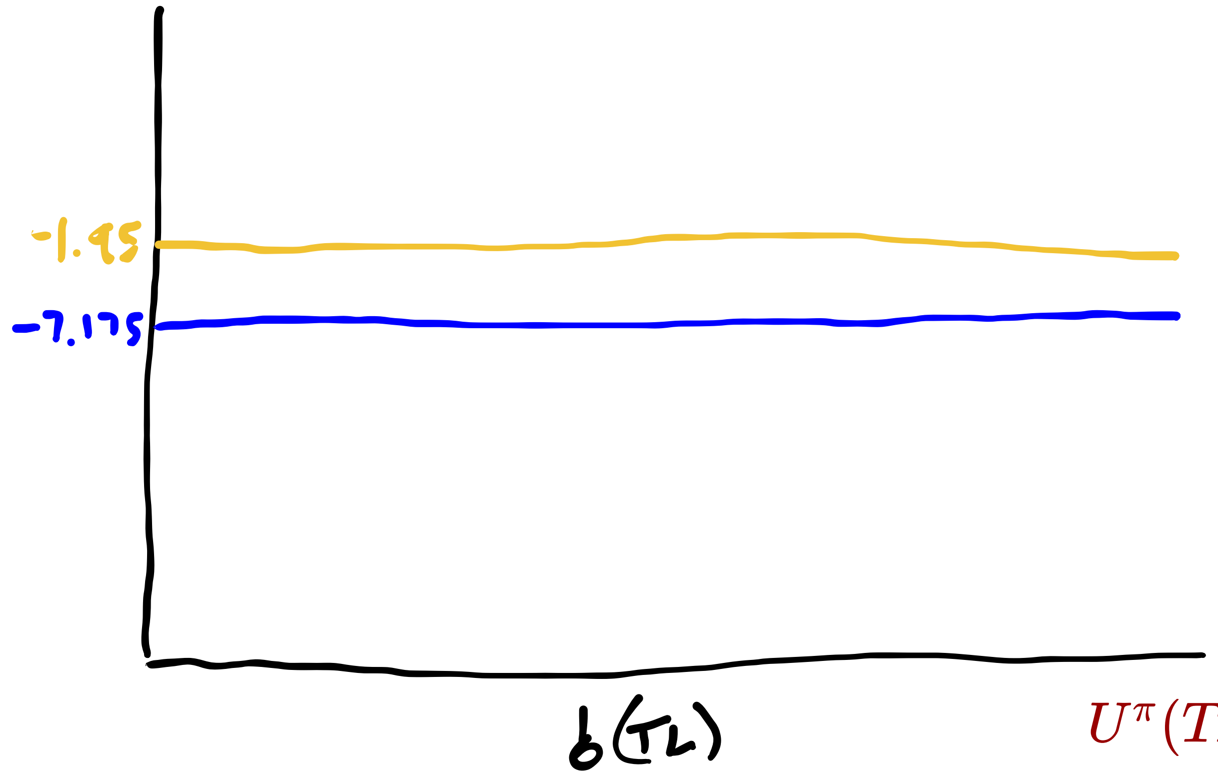

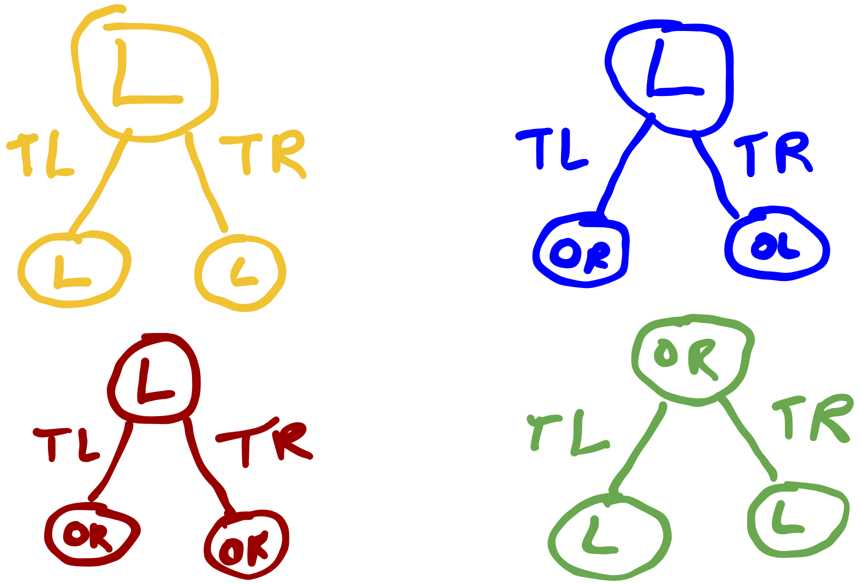

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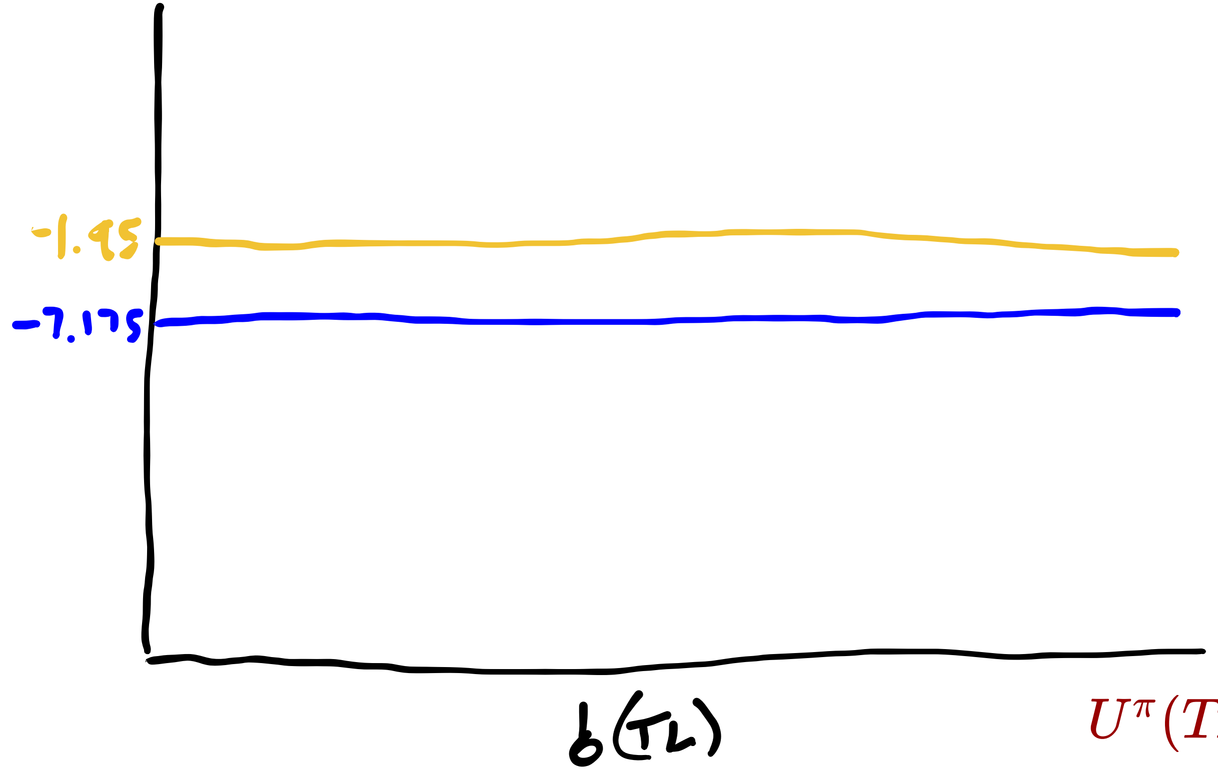

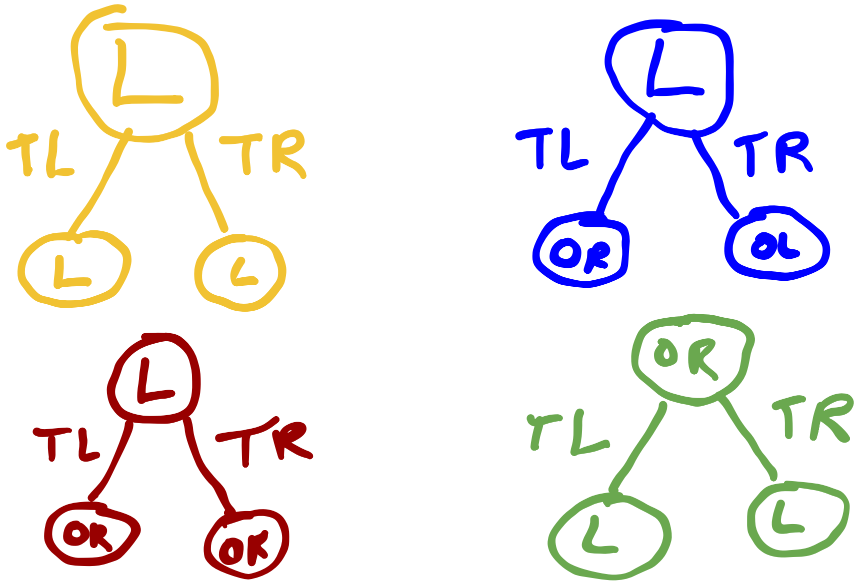

$U^{\pi}(TL) = -1 + \gamma\, 10$

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$U^{\pi}(TL) = 10 + \gamma(-1)$

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Alpha Vectors for Conditional Plans

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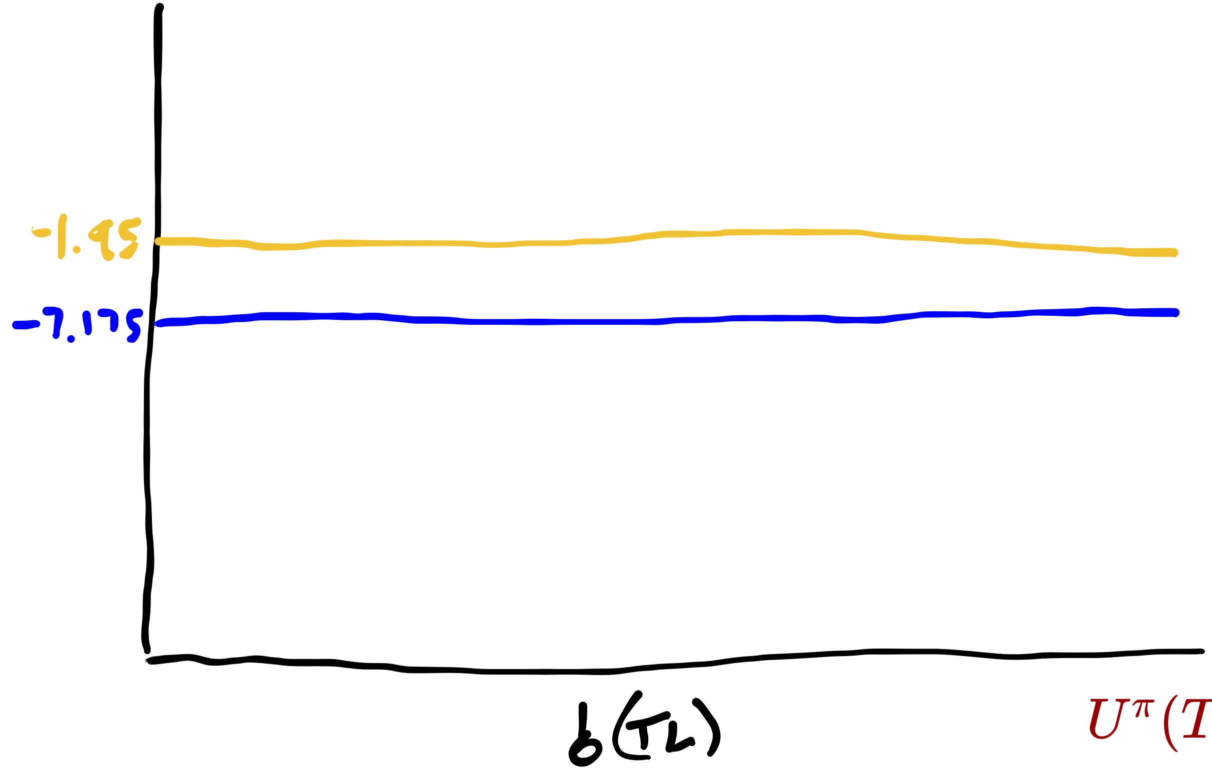

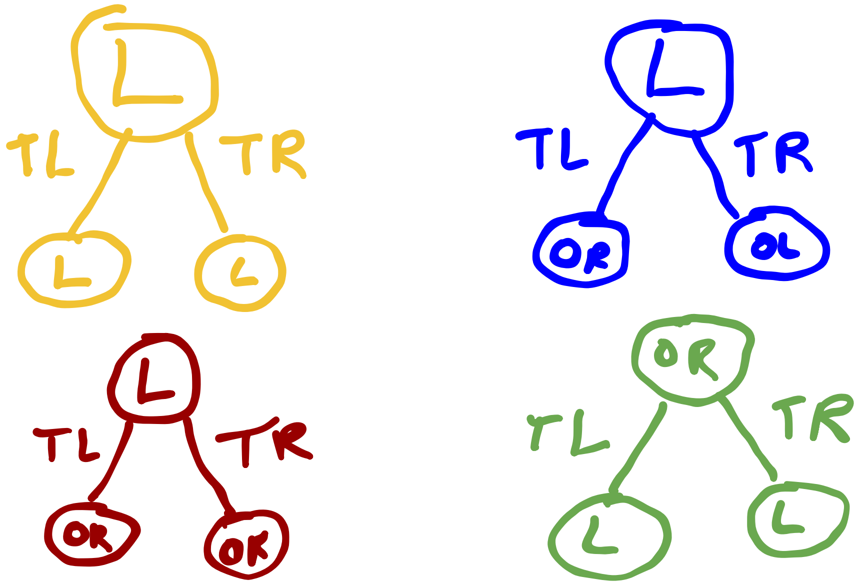

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Alpha Vectors for Conditional Plans

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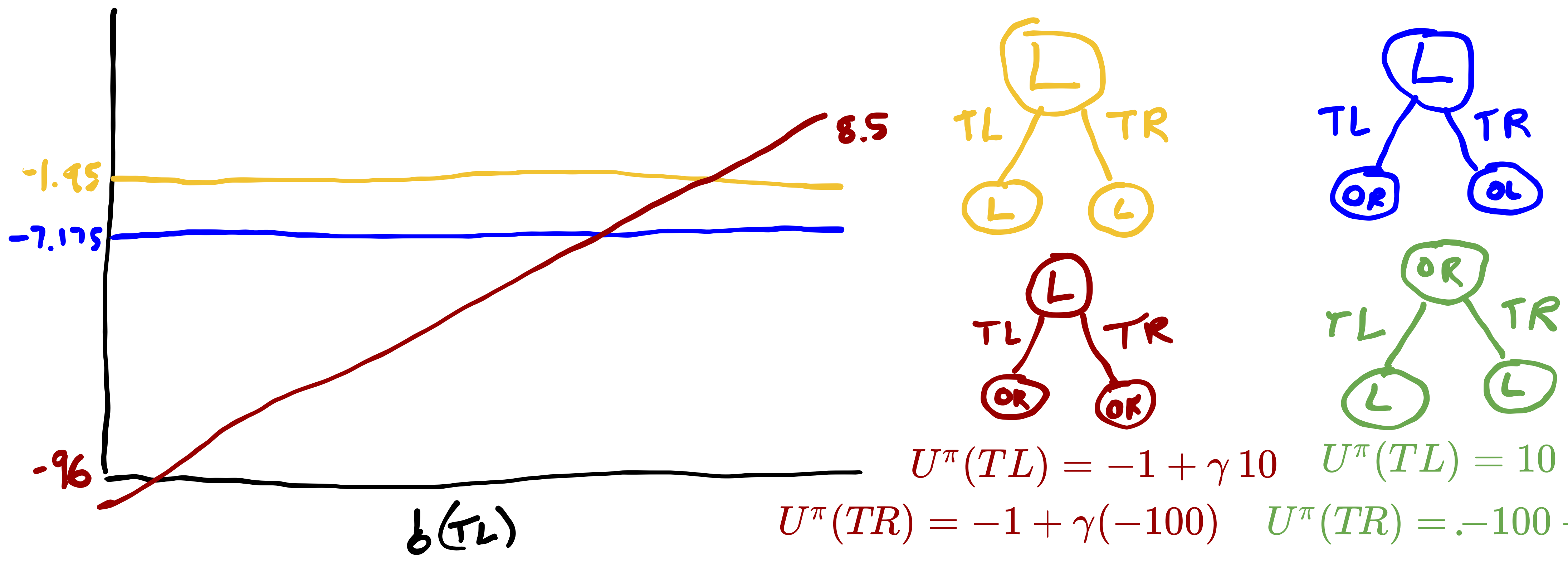

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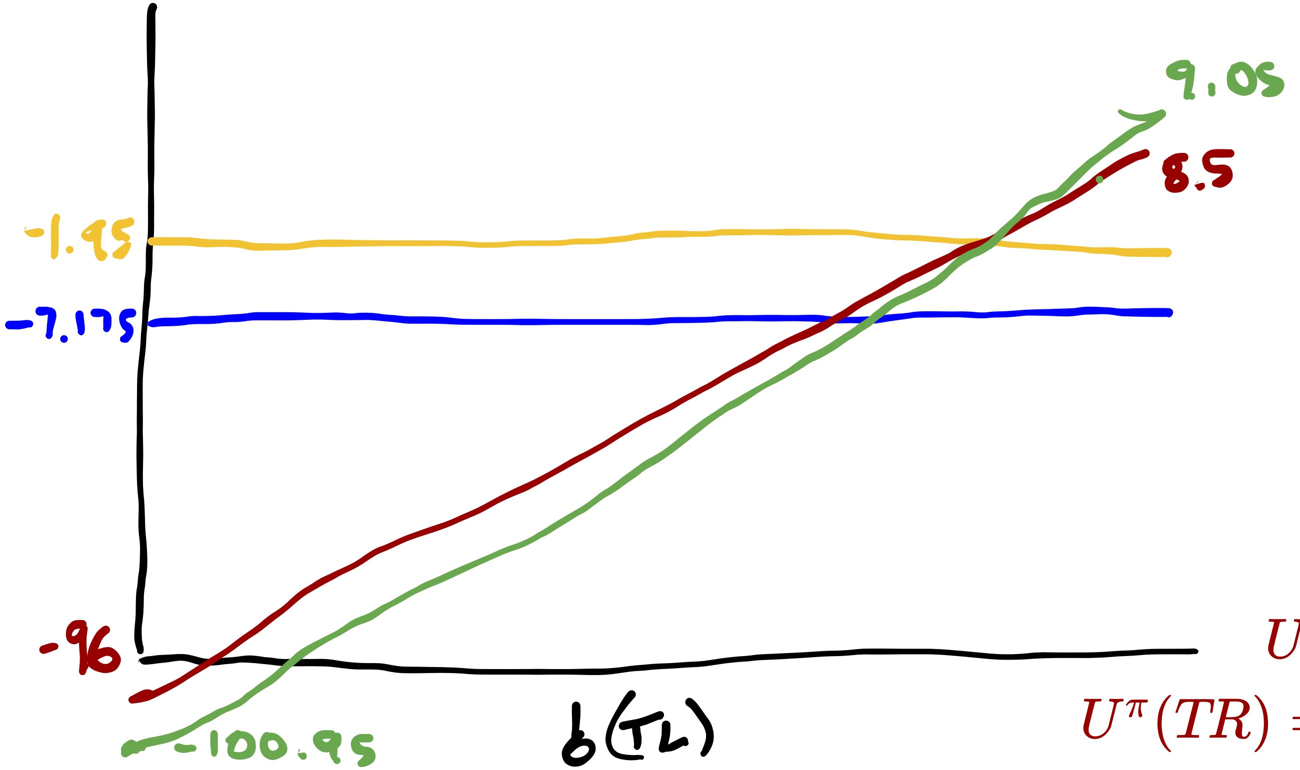

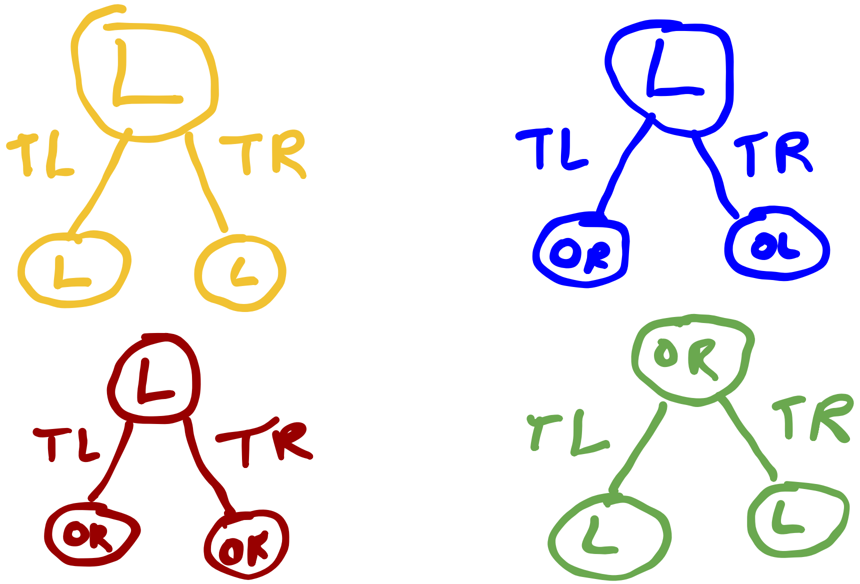

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POMDP Value Functions

POMDP Value Functions

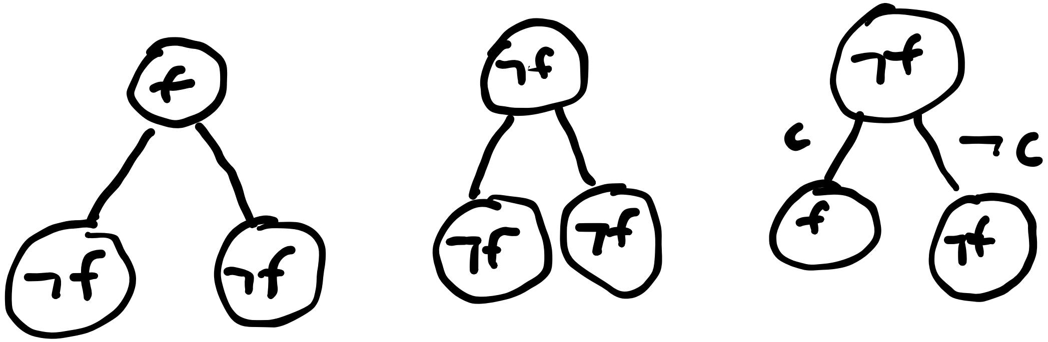
$$V^*(b) = \max_{\alpha \in \Gamma} \alpha^\top b$$

Exercise: 2-Step Crying Baby α Vectors

$$S = \{h, \neg h\} \qquad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \qquad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \qquad T(\neg h \mid \cdot, f) = 1.0$$



$$R(s,a) = R(s) + R(a)$$

$$U^{\pi}(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_{o} O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

$$R(s) = \begin{cases} -10 \text{ if } s = h \\ 0 \text{ otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 \text{ if } a = f \\ 0 \text{ otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8)$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$

α-Vector Pruning

Alpha Vector Expansion

POMDP Value Iteration (horizon d)

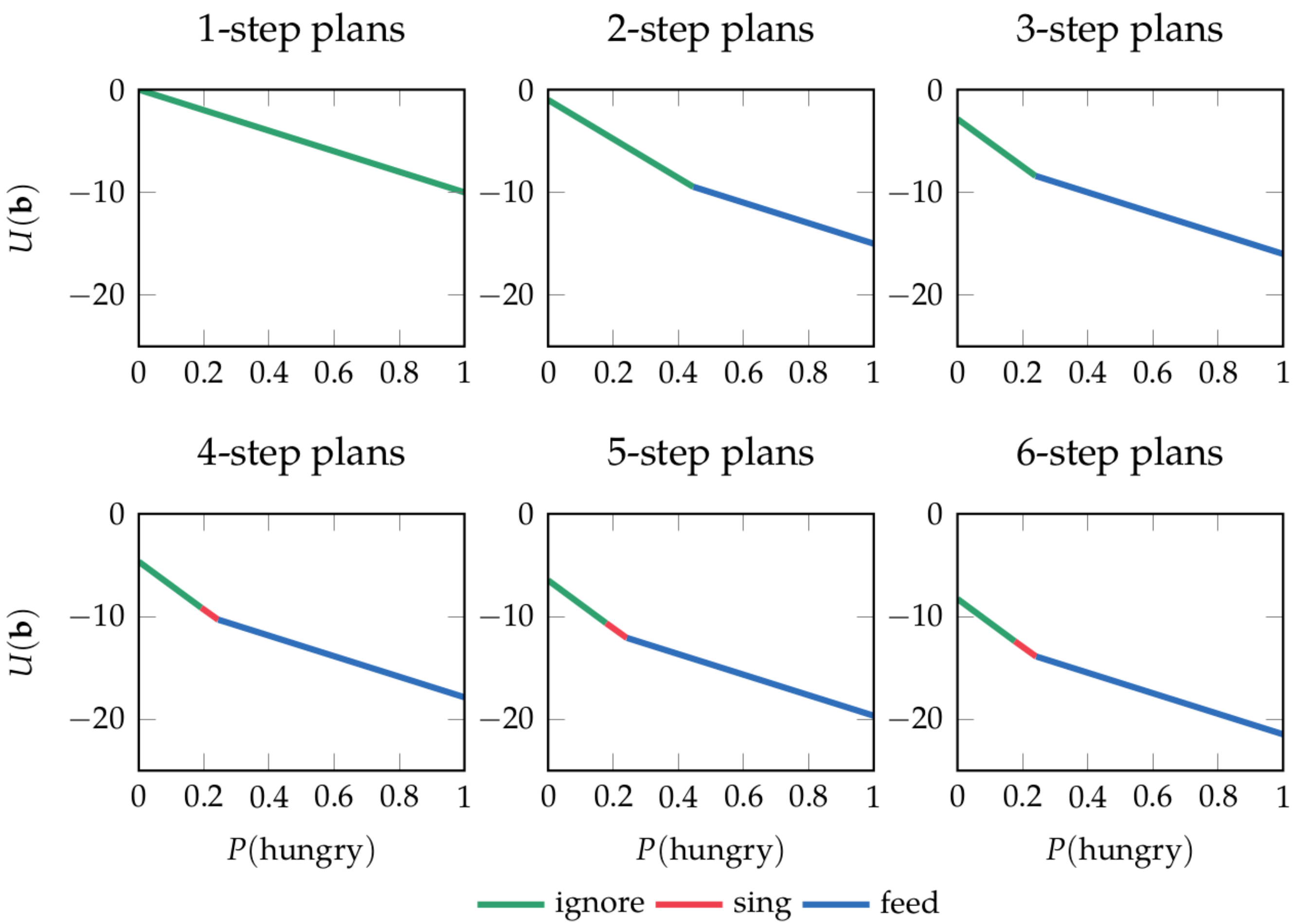
$\Gamma^0 \leftarrow \emptyset$

for $n \in 1 \ldots d$

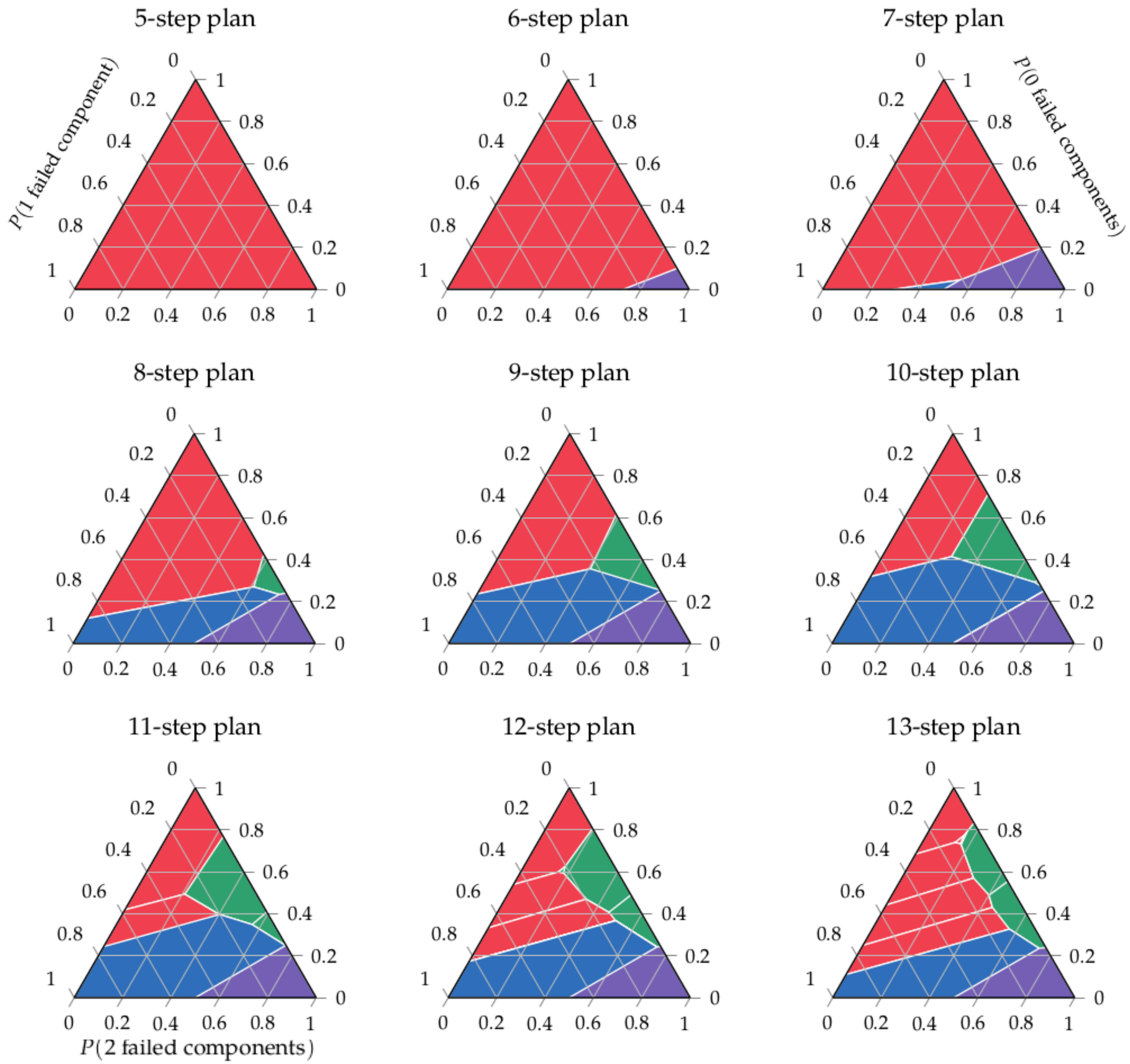
    Construct Γ^n by expanding with Γ^{n-1}

    Prune Γ^n

Finite Horizon POMDP Value Iteration



Finite Horizon POMDP Value Iteration



Recap

Recap

- A POMDP is an MDP on the ____________

Recap

- A POMDP is an MDP on the belief space

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of ________________

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α-vectors

Recap

- A POMDP is an MDP on the <u>belief space</u>
- The value function of a discrete POMDP can be represented by a set of <u>α-vectors</u>
- Each α vector corresponds to a ________________

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α-vectors
- Each α vector corresponds to a conditional plan